

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

research@deepseek.com

Abstract

We introduce our first-generation reasoning models, DeepSeek-R1-Zero and DeepSeek-R1. DeepSeek-R1-Zero, a model trained via large-scale reinforcement learning (RL) without supervised fine-tuning (SFT) as a preliminary step, demonstrates remarkable reasoning capabilities. Through RL, DeepSeek-R1-Zero naturally emerges with numerous powerful and intriguing reasoning behaviors. However, it encounters challenges such as poor readability, and language mixing. To address these issues and further enhance reasoning performance, we introduce DeepSeek-R1, which incorporates multi-stage training and cold-start data before RL. DeepSeek-R1 achieves performance comparable to OpenAI-o1-1217 on reasoning tasks. To support the research community, we open-source DeepSeek-R1-Zero, DeepSeek-R1, and six dense models (1.5B, 7B, 8B, 14B, 32B, 70B) distilled from DeepSeek-R1 based on Qwen and Llama.

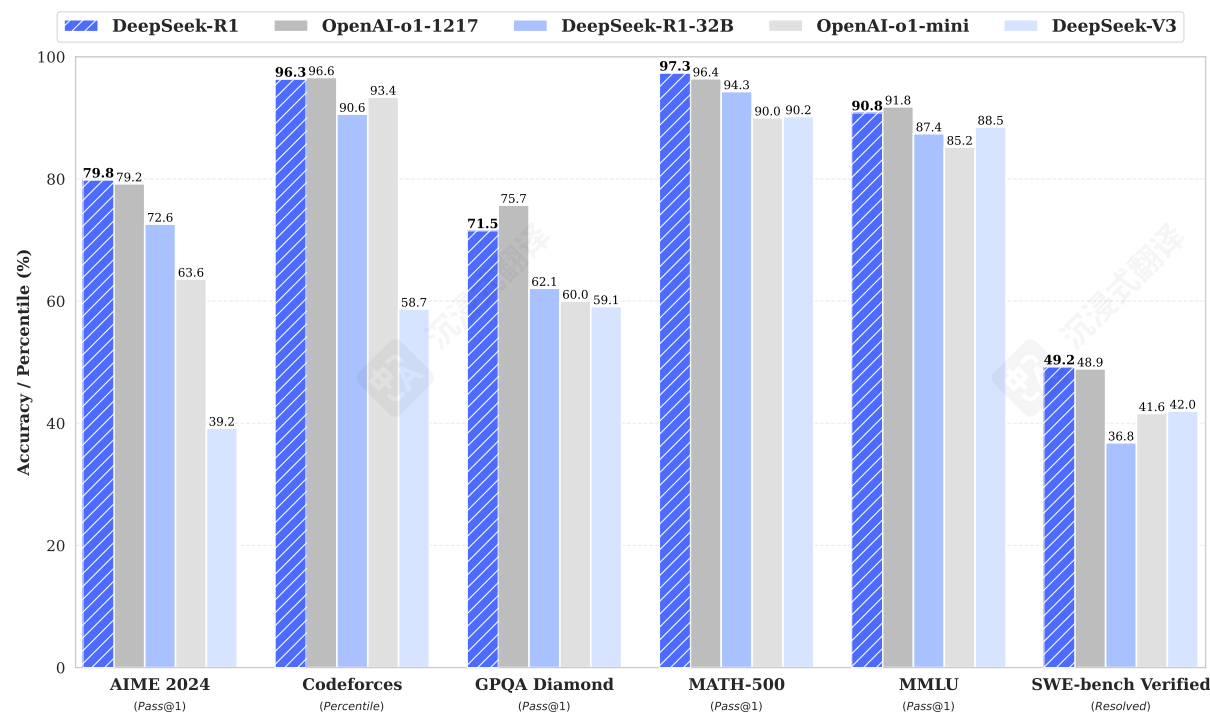


Figure 1 | Benchmark performance of DeepSeek-R1.

DeepSeek-R1: 通过强化学习激励大型语言模型的推理能力

DeepSeek-AI

research@deepseek.com

摘要

我们介绍了我们第一代推理模型，DeepSeek-R1-Zero 和 DeepSeek-R1。DeepSeek-R1-Zero 是一个通过大规模强化学习（RL）训练的模型，没有作为初步步骤进行监督微调（SFT），展示了卓越的推理能力。通过 RL，DeepSeek-R1-Zero 自然地涌现出许多强大且有趣的推理行为。然而，它面临诸如可读性差和语言混合等挑战。为了解决这些问题并进一步提高推理性能，我们引入了 DeepSeek-R1，它结合了多阶段训练和强化学习前的冷启动数据。DeepSeek-R1 在推理任务上的性能与 OpenAI-o1-1217 相当。为了支持研究社区，我们开源了 DeepSeek-R1-Zero、DeepSeek-R1 和六个从 DeepSeek-R1 基于Qwen和Llama蒸馏出的密集模型（1.5B、7B、8B、14B、32B、70B）。

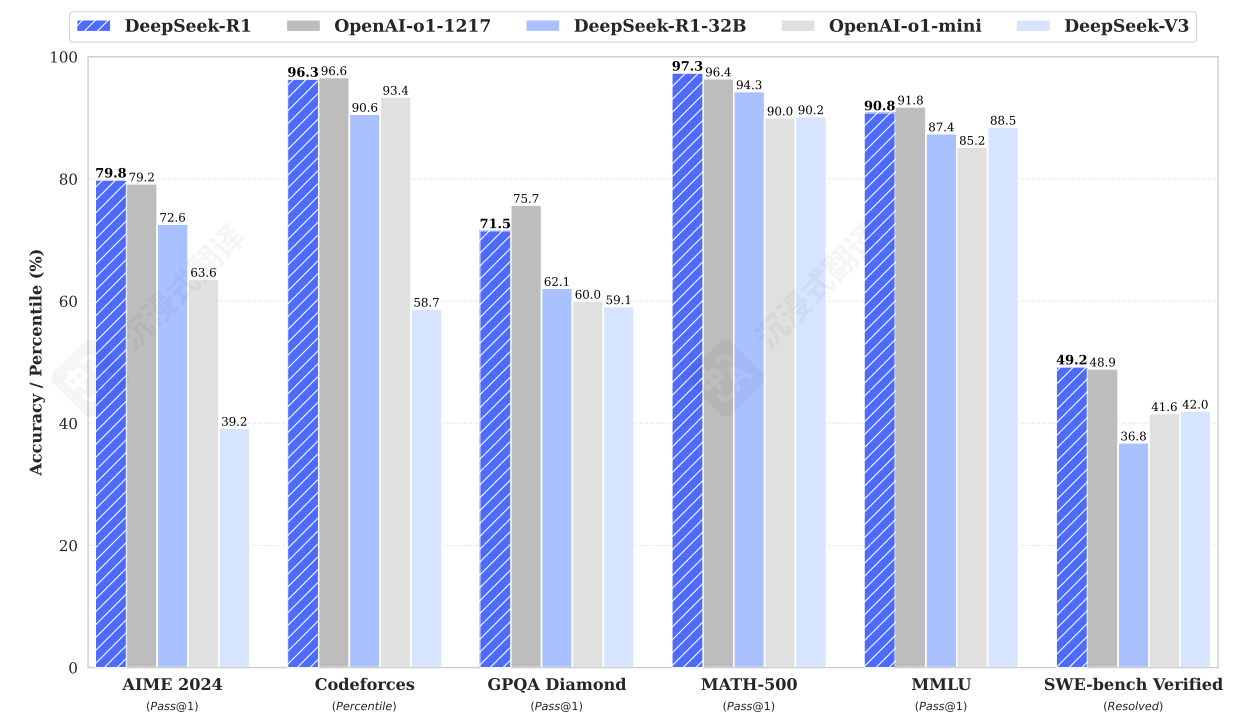


图 1 | DeepSeek-R1基准性能。

Contents

1 Introduction 3

1.1 Contributions 4

1.2 Summary of Evaluation Results 4

2 Approach 5

2.1 Overview 5

2.2 DeepSeek-R1-Zero: Reinforcement Learning on the Base Model 5

2.2.1 Reinforcement Learning Algorithm 5

2.2.2 Reward Modeling 6

2.2.3 Training Template 6

2.2.4 Performance, Self-evolution Process and Aha Moment of DeepSeek-R1-Zero 6

2.3 DeepSeek-R1: Reinforcement Learning with Cold Start 9

2.3.1 Cold Start 9

2.3.2 Reasoning-oriented Reinforcement Learning 10

2.3.3 Rejection Sampling and Supervised Fine-Tuning 10

2.3.4 Reinforcement Learning for all Scenarios 11

2.4 Distillation: Empower Small Models with Reasoning Capability 11

3 Experiment 11

3.1 DeepSeek-R1 Evaluation 13

3.2 Distilled Model Evaluation 14

4 Discussion 14

4.1 Distillation v.s. Reinforcement Learning 14

4.2 Unsuccessful Attempts 15

5 Conclusion, Limitations, and Future Work 16

A Contributions and Acknowledgments 20

内容

1 简介 31.1 贡献.4 1.2 评估结果摘要.42 方法 52.1 概述.5 2.2 DeepSeek-R1-Zero: 在基础模型上进行强化学习.5 2.2.1 强化学习算法.5 2.2.2 奖励建模.6 2.2.3 训练模板.6 2.2.4 DeepSeek-R1-Zero 的性能、自我进化过程和“啊哈时刻” 6 2.3 DeepSeek-R1: 带冷启动的强化学习.9 2.3.1 冷启动.9 2.3.2 以推理为导向的强化学习.10 2.3.3 拒绝采样和监督微调.10 2.3.4 全场景强化学习.11 2.4 蒸馏: 赋予小模型推理能力.

1. Introduction

In recent years, Large Language Models (LLMs) have been undergoing rapid iteration and evolution (Anthropic, 2024; Google, 2024; OpenAI, 2024a), progressively diminishing the gap towards Artificial General Intelligence (AGI).

Recently, post-training has emerged as an important component of the full training pipeline. It has been shown to enhance accuracy on reasoning tasks, align with social values, and adapt to user preferences, all while requiring relatively minimal computational resources against pre-training. In the context of reasoning capabilities, OpenAI’s o1 (OpenAI, 2024b) series models were the first to introduce inference-time scaling by increasing the length of the Chain-of-Thought reasoning process. This approach has achieved significant improvements in various reasoning tasks, such as mathematics, coding, and scientific reasoning. However, the challenge of effective test-time scaling remains an open question for the research community. Several prior works have explored various approaches, including process-based reward models (Lightman et al., 2023; Uesato et al., 2022; Wang et al., 2023), reinforcement learning (Kumar et al., 2024), and search algorithms such as Monte Carlo Tree Search and Beam Search (Feng et al., 2024; Trinh et al., 2024; Xin et al., 2024). However, none of these methods has achieved general reasoning performance comparable to OpenAI’s o1 series models.

In this paper, we take the first step toward improving language model reasoning capabilities using pure reinforcement learning (RL). Our goal is to explore the potential of LLMs to develop reasoning capabilities without any supervised data, focusing on their self-evolution through a pure RL process. Specifically, we use DeepSeek-V3-Base as the base model and employ GRPO (Shao et al., 2024) as the RL framework to improve model performance in reasoning. During training, DeepSeek-R1-Zero naturally emerged with numerous powerful and interesting reasoning behaviors. After thousands of RL steps, DeepSeek-R1-Zero exhibits super performance on reasoning benchmarks. For instance, the pass@1 score on AIME 2024 increases from 15.6% to 71.0%, and with majority voting, the score further improves to 86.7%, matching the performance of OpenAI-o1-0912.

However, DeepSeek-R1-Zero encounters challenges such as poor readability, and language mixing. To address these issues and further enhance reasoning performance, we introduce DeepSeek-R1, which incorporates a small amount of cold-start data and a multi-stage training pipeline. Specifically, we begin by collecting thousands of cold-start data to fine-tune the DeepSeek-V3-Base model. Following this, we perform reasoning-oriented RL like DeepSeek-R1-Zero. Upon nearing convergence in the RL process, we create new SFT data through rejection sampling on the RL checkpoint, combined with supervised data from DeepSeek-V3 in domains such as writing, factual QA, and self-cognition, and then retrain the DeepSeek-V3-Base model. After fine-tuning with the new data, the checkpoint undergoes an additional RL process, taking into account prompts from all scenarios. After these steps, we obtained a checkpoint referred to as DeepSeek-R1, which achieves performance on par with OpenAI-o1-1217.

We further explore distillation from DeepSeek-R1 to smaller dense models. Using Qwen2.5-32B (Qwen, 2024b) as the base model, direct distillation from DeepSeek-R1 outperforms applying RL on it. This demonstrates that the reasoning patterns discovered by larger base models are crucial for improving reasoning capabilities. We open-source the distilled Qwen and Llama (Dubey et al., 2024) series. Notably, our distilled 14B model outperforms state-of-the-art open-source QwQ-32B-Preview (Qwen, 2024a) by a large margin, and the distilled 32B and 70B models set a new record on the reasoning benchmarks among dense models.

1. 简介

近年来，大型语言模型（LLMs）正在经历快速迭代和演进（Anthropic, 2024; Google, 2024; OpenAI, 2024a），逐步缩小与通用人工智能（AGI）的差距。

最近，后训练已成为完整训练流程中的一个重要组成部分。研究表明，它在推理任务上提高了准确性，与社会价值观保持一致，并适应用户偏好，同时相对于预训练，所需的计算资源相对较少。在推理能力方面，OpenAI的o1（OpenAI, 2024b）系列模型率先引入了推理时扩展，通过增加思维链（Chain-of-Thought）推理过程的长度来实现。这种方法在各种推理任务中取得了显著改进，例如数学、编程和科学推理。然而，有效测试时扩展的挑战仍然是研究界的一个开放性问题。已有一些研究探索了各种方法，包括基于过程的奖励模型（Lightman等人, 2023; Uesato等人, 2022; Wang等人, 2023）、强化学习（Kumar等人, 2024），以及蒙特卡洛树搜索和波束搜索等搜索算法（Feng等人, 2024; Trinh等人, 2024; Xin等人, 2024）。然而，这些方法中还没有一种实现了与OpenAI的o1系列模型相当的一般推理性能。

在本文中，我们迈出了使用纯强化学习（RL）提升语言模型推理能力的第一步。我们的目标是探索LLM在没有任何监督数据的情况下发展推理能力的潜力，重点关注通过纯RL过程进行的自我进化。具体来说，我们使用DeepSeek-V3-Base作为基础模型，并采用GRPO（Shao等人, 2024年）作为RL框架来提升模型在推理方面的性能。在训练过程中，DeepSeek-R1-Zero自然涌现出许多强大且有趣的推理行为。经过数千次RL步骤后，DeepSeek-R1-Zero在推理基准测试中表现出超强的性能。例如，AIME 2024的pass@1得分从15.6%提高到71.0%，通过多数投票，得分进一步提高到86.7%，与OpenAI-o1-0912的性能相当。

然而，DeepSeek-R1-Zero遇到了可读性差、语言混杂等挑战。为了解决这些问题并进一步提升推理性能，我们引入了DeepSeek-R1，该模型结合了少量冷启动数据和分阶段训练流程。具体来说，我们首先收集数千个冷启动数据来微调DeepSeek-V3-Base模型。随后，我们执行类似DeepSeek-R1-Zero的推理导向RL。在RL过程接近收敛时，我们通过在RL检查点上拒绝采样创建新的SFT数据，并结合来自DeepSeek-V3的监督数据（涵盖写作、事实性问答和自我认知等领域），然后重新训练DeepSeek-V3-Base模型。在用新数据微调后，检查点会经历额外的RL过程，考虑所有场景中的提示。完成这些步骤后，我们获得了一个称为DeepSeek-R1的检查点，其性能与OpenAI-o1-1217相当。

我们进一步探索从DeepSeek-R1向更小的密集模型蒸馏。以Qwen2.5-32B（Qwen, 2024b）作为基础模型，直接从DeepSeek-R1进行蒸馏的表现优于对其应用强化学习。这表明大型基础模型发现的推理模式对于提升推理能力至关重要。我们开源了蒸馏后的Qwen和Llama（Dubey et al., 2024）系列。值得注意的是，我们蒸馏的14B模型大幅超越了最先进的开源QwQ-32B-Preview（Qwen, 2024a），而蒸馏的32B和70B模型在密集模型的推理基准测试中创下了新纪录。

1.1. Contributions

Post-Training: Large-Scale Reinforcement Learning on the Base Model

- We directly apply RL to the base model without relying on supervised fine-tuning (SFT) as a preliminary step. This approach allows the model to explore chain-of-thought (CoT) for solving complex problems, resulting in the development of DeepSeek-R1-Zero. DeepSeek-R1-Zero demonstrates capabilities such as self-verification, reflection, and generating long CoTs, marking a significant milestone for the research community. Notably, it is the first open research to validate that reasoning capabilities of LLMs can be incentivized purely through RL, without the need for SFT. This breakthrough paves the way for future advancements in this area.
- We introduce our pipeline to develop DeepSeek-R1. The pipeline incorporates two RL stages aimed at discovering improved reasoning patterns and aligning with human preferences, as well as two SFT stages that serve as the seed for the model's reasoning and non-reasoning capabilities. We believe the pipeline will benefit the industry by creating better models.

Distillation: Smaller Models Can Be Powerful Too

- We demonstrate that the reasoning patterns of larger models can be distilled into smaller models, resulting in better performance compared to the reasoning patterns discovered through RL on small models. The open source DeepSeek-R1, as well as its API, will benefit the research community to distill better smaller models in the future.
- Using the reasoning data generated by DeepSeek-R1, we fine-tuned several dense models that are widely used in the research community. The evaluation results demonstrate that the distilled smaller dense models perform exceptionally well on benchmarks. DeepSeek-R1-Distill-Qwen-7B achieves 55.5% on AIME 2024, surpassing QwQ-32B-Preview. Additionally, DeepSeek-R1-Distill-Qwen-32B scores 72.6% on AIME 2024, 94.3% on MATH-500, and 57.2% on LiveCodeBench. These results significantly outperform previous open-source models and are comparable to o1-mini. We open-source distilled 1.5B, 7B, 8B, 14B, 32B, and 70B checkpoints based on Qwen2.5 and Llama3 series to the community.

1.2. Summary of Evaluation Results

- **Reasoning tasks:** (1) DeepSeek-R1 achieves a score of 79.8% Pass@1 on AIME 2024, slightly surpassing OpenAI-o1-1217. On MATH-500, it attains an impressive score of 97.3%, performing on par with OpenAI-o1-1217 and significantly outperforming other models. (2) On coding-related tasks, DeepSeek-R1 demonstrates expert level in code competition tasks, as it achieves 2,029 Elo rating on Codeforces outperforming 96.3% human participants in the competition. For engineering-related tasks, DeepSeek-R1 performs slightly better than DeepSeek-V3, which could help developers in real world tasks.
- **Knowledge:** On benchmarks such as MMLU, MMLU-Pro, and GPQA Diamond, DeepSeek-R1 achieves outstanding results, significantly outperforming DeepSeek-V3 with scores of 90.8% on MMLU, 84.0% on MMLU-Pro, and 71.5% on GPQA Diamond. While its performance is slightly below that of OpenAI-o1-1217 on these benchmarks, DeepSeek-R1 surpasses other closed-source models, demonstrating its competitive edge in educational tasks. On the factual benchmark SimpleQA, DeepSeek-R1 outperforms DeepSeek-V3, demonstrating its capability in handling fact-based queries. A similar trend is observed where OpenAI-o1 surpasses 4o on this benchmark.

1.1. 贡献

训练后：在基础模型上进行大规模强化学习

- 我们直接将强化学习（RL）应用于基础模型，而不依赖于作为初步步骤的监督微调（SFT）。这种方法允许模型探索思维链（CoT）以解决复杂问题，从而开发出 DeepSeek-R1-Zero。DeepSeek-R1-Zero 展示了自我验证、反思以及生成思维链等能力，标志着研究社区的重大里程碑。值得注意的是，它是首个公开研究验证 LLM 的推理能力可以通过纯强化学习激励，而无需 SFT。这一突破为该领域的未来进步铺平了道路。
- 我们介绍了开发 DeepSeek-R1 的流程。该流程包含两个强化学习阶段，旨在发现改进的推理模式并与人类偏好对齐，以及两个监督微调阶段，作为模型推理和非推理能力的种子。我们相信该流程将通过创建更好的模型使行业受益。

蒸馏：小型模型也能很强大

- 我们证明了大型模型的推理模式可以被蒸馏到小型模型中，其性能优于通过 RL 在小型模型发现的推理模式。开源的 DeepSeek-R1 及其 API 将有助于研究社区在未来蒸馏出更好的小型模型。
- 使用 DeepSeek-R1 生成的推理数据，我们微调了研究社区广泛使用的几个密集模型。评估结果表明，蒸馏后的小型密集模型在基准测试上表现优异。DeepSeek-R1-Distill-Qwen-7B 在 AIME 2024 上达到 55.5%，超过 QwQ-32B-Preview。此外，DeepSeek-R1-Distill-Qwen-32B 在 AIME2024 上得分 72.6%，在 MATH-500 上得分 94.3%，在 LiveCodeBench 上得分 57.2%。这些结果显著优于之前的开源模型，并与 o1-mini 相当。我们基于 Qwen2.5 和 Llama3 系列开源了蒸馏后的 1.5B、7B、8B、14B、32B 和 70B 检查点，供社区使用。

1.2. 评估结果摘要

- **推理任务:** (1) DeepSeek-R1 在 AIME2024 上取得了 79.8% 的 Pass@1 得分，略微超过 OpenAI-o1-1217。在 MATH-500 上，它达到了令人印象深刻的 97.3% 得分，表现与 OpenAI-o1-1217 相当，并显著优于其他模型。(2) 在与编程相关的任务中，DeepSeek-R1 在代码竞赛任务中表现出专家水平，它在 Codeforces 上获得了 2,029 Elo 评分，超过了 96.3% 的参赛者。在工程相关任务中，DeepSeek-R1 的表现略优于 DeepSeek-V3，这有助于开发者在实际任务中。
- **知识:** 在 MMLU、MMLU-Pro 和 GPQA Diamond 等基准测试中，DeepSeek-R1 取得了优异的成绩，显著优于 DeepSeek-V3，在 MMLU 上得分为 90.8%，在 MMLU-Pro 上得分为 84.0%，在 GPQA Diamond 上得分为 71.5%。虽然它在这些基准测试上的表现略低于 OpenAI-o1-1217，但 DeepSeek-R1 超过了其他闭源模型，在教育任务中展现出其竞争优势。在事实基准测试 SimpleQA 上，DeepSeek-R1 优于 DeepSeek-V3，展示了其处理基于事实查询的能力。在这个基准测试上，OpenAI-o1 也超过了 4o，呈现出类似的趋势。

- **Others:** DeepSeek-R1 also excels in a wide range of tasks, including creative writing, general question answering, editing, summarization, and more. It achieves an impressive length-controlled win-rate of 87.6% on AlpacaEval 2.0 and a win-rate of 92.3% on ArenaHard, showcasing its strong ability to intelligently handle non-exam-oriented queries. Additionally, DeepSeek-R1 demonstrates outstanding performance on tasks requiring long-context understanding, substantially outperforming DeepSeek-V3 on long-context benchmarks.

2. Approach

2.1. Overview

Previous work has heavily relied on large amounts of supervised data to enhance model performance. In this study, we demonstrate that reasoning capabilities can be significantly improved through large-scale reinforcement learning (RL), even without using supervised fine-tuning (SFT) as a cold start. Furthermore, performance can be further enhanced with the inclusion of a small amount of cold-start data. In the following sections, we present: (1) DeepSeek-R1-Zero, which applies RL directly to the base model without any SFT data, and (2) DeepSeek-R1, which applies RL starting from a checkpoint fine-tuned with thousands of long Chain-of-Thought (CoT) examples. 3) Distill the reasoning capability from DeepSeek-R1 to small dense models.

2.2. DeepSeek-R1-Zero: Reinforcement Learning on the Base Model

Reinforcement learning has demonstrated significant effectiveness in reasoning tasks, as evidenced by our previous works (Shao et al., 2024; Wang et al., 2023). However, these works heavily depended on supervised data, which are time-intensive to gather. In this section, we explore the potential of LLMs to develop reasoning capabilities **without any supervised data**, focusing on their self-evolution through a pure reinforcement learning process. We start with a brief overview of our RL algorithm, followed by the presentation of some exciting results, and hope this provides the community with valuable insights.

2.2.1. Reinforcement Learning Algorithm

Group Relative Policy Optimization In order to save the training costs of RL, we adopt Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which foregoes the critic model that is typically the same size as the policy model, and estimates the baseline from group scores instead. Specifically, for each question q , GRPO samples a group of outputs $\{o_1, o_2, \dots, o_G\}$ from the old policy $\pi_{\theta_{old}}$ and then optimizes the policy model π_{θ} by maximizing the following objective:

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right), \quad (1)$$

$$\mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1, \quad (2)$$

where ϵ and β are hyper-parameters, and A_i is the advantage, computed using a group of rewards $\{r_1, r_2, \dots, r_G\}$ corresponding to the outputs within each group:

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}. \quad (3)$$

- **Others:** DeepSeek-R1 在多种任务上表现出色，包括创意写作、一般问答、编辑、摘要等。它在 AlpacaEval 2.0 上实现了 87.6% 的长度控制胜率，在 ArenaHard 上实现了 92.3% 的胜率，展示了其智能处理非考试导向查询的强大能力。此外，DeepSeek-R1 在需要长上下文理解的任务上表现出色，在长上下文基准测试中大幅优于 DeepSeek-V3。

2. 方法

2.1. 概述

以往工作严重依赖大量监督数据来提升模型性能。在本研究中，我们证明通过大规模强化学习（RL），即使不使用监督微调（SFT）作为冷启动，推理能力也能得到显著提升。此外，通过加入少量冷启动数据，性能可以进一步提升。在以下部分，我们介绍了：(1) DeepSeek-R1-Zero，它直接将 RL 应用于基础模型，不使用任何 SFT 数据，以及 (2) DeepSeek-R1，它从使用数千个长思维链（CoT）示例微调的检查点开始应用 RL。3) 将 DeepSeek-R1 的推理能力蒸馏到小型密集模型中。

2.2. DeepSeek-R1-Zero: 基于基础模型的强化学习

强化学习在推理任务中已展现出显著的有效性，这一点由我们之前的工作（Shao et al., 2024; Wang et al., 2023）所证实。然而，这些工作严重依赖于监督数据，而收集这些数据需要耗费大量时间。在本节中，我们探索了大型语言模型（LLMs）在开发推理能力方面的潜力 **而无需任何监督数据**，重点关注它们通过纯粹的强化学习过程进行自我进化的能力。我们从简要概述我们的强化学习（RL）算法开始，然后介绍一些令人兴奋的结果，希望这能为社区提供有价值的见解。

2.2.1. 强化学习算法

组相对策略优化 为了节省强化学习（RL）的训练成本，我们采用了组相对策略优化（GRPO）（Shao et al., 2024），该优化方法放弃了通常与策略模型规模相同的评价者模型，而是从组分分数中估计基线。具体来说，对于每个问题 q ，GRPO 从旧策略 $\pi_{\theta_{old}}$ 中采样一组输出 $\{o_1, o_2, \dots, o_G\}$ ，然后通过最大化以下目标来优化策略模型 π_{θ} ：

$$\mathcal{J}_{GRPO}(\theta) = \mathbb{E}[q \sim P(Q), \{o_i\}_{i=1}^G \sim \pi_{\theta_{old}}(O|q)] \frac{1}{G} \sum_{i=1}^G \left(\min \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_{\theta}(o_i|q)}{\pi_{\theta_{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) \right), \quad (1)$$

$$\mathbb{D}_{KL}(\pi_{\theta} || \pi_{ref}) = \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - \log \frac{\pi_{ref}(o_i|q)}{\pi_{\theta}(o_i|q)} - 1, \quad (2)$$

在 ϵ 和 β 是超参数，而 A_i 是优势，使用对应于每个组中的输出的奖励组 $\{r_1, r_2, \dots, r_G\}$ 计算得出：

$$A_i = \frac{r_i - \text{mean}(\{r_1, r_2, \dots, r_G\})}{\text{std}(\{r_1, r_2, \dots, r_G\})}. \quad (3)$$

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within <think> </think> and <answer> </answer> tags, respectively, i.e., <think> reasoning process here </think> <answer> answer here </answer>. User: **prompt**. Assistant:

Table 1 | Template for DeepSeek-R1-Zero. **prompt** will be replaced with the specific reasoning question during training.

2.2.2. Reward Modeling

The reward is the source of the training signal, which decides the optimization direction of RL. To train DeepSeek-R1-Zero, we adopt a rule-based reward system that mainly consists of two types of rewards:

- **Accuracy rewards:** The accuracy reward model evaluates whether the response is correct. For example, in the case of math problems with deterministic results, the model is required to provide the final answer in a specified format (e.g., within a box), enabling reliable rule-based verification of correctness. Similarly, for LeetCode problems, a compiler can be used to generate feedback based on predefined test cases.
- **Format rewards:** In addition to the accuracy reward model, we employ a format reward model that enforces the model to put its thinking process between ‘<think>’ and ‘</think>’ tags.

We do not apply the outcome or process neural reward model in developing DeepSeek-R1-Zero, because we find that the neural reward model may suffer from reward hacking in the large-scale reinforcement learning process, and retraining the reward model needs additional training resources and it complicates the whole training pipeline.

2.2.3. Training Template

To train DeepSeek-R1-Zero, we begin by designing a straightforward template that guides the base model to adhere to our specified instructions. As depicted in Table 1, this template requires DeepSeek-R1-Zero to first produce a reasoning process, followed by the final answer. We intentionally limit our constraints to this structural format, avoiding any content-specific biases—such as mandating reflective reasoning or promoting particular problem-solving strategies—to ensure that we can accurately observe the model’s natural progression during the RL process.

2.2.4. Performance, Self-evolution Process and Aha Moment of DeepSeek-R1-Zero

Performance of DeepSeek-R1-Zero Figure 2 depicts the performance trajectory of DeepSeek-R1-Zero on the AIME 2024 benchmark throughout the RL training process. As illustrated, DeepSeek-R1-Zero demonstrates a steady and consistent enhancement in performance as the RL training advances. Notably, the average pass@1 score on AIME 2024 shows a significant increase, jumping from an initial 15.6% to an impressive 71.0%, reaching performance levels comparable to OpenAI-o1-0912. This significant improvement highlights the efficacy of our RL algorithm in optimizing the model’s performance over time.

Table 2 provides a comparative analysis between DeepSeek-R1-Zero and OpenAI’s o1-0912 models across a variety of reasoning-related benchmarks. The findings reveal that RL empowers

用户与助手之间的对话。用户提出问题，助手解答。助手首先在心中思考推理过程，然后向用户提供答案。推理过程和答案分别用 <思考> </思考> 和<回答> </回答> 标签表示，即 <思考>推理过程在这里 </思考><回答> 答案在这里 </回答>。用户： **提示**。助手：

表格 1 | DeepSeek-R1-Zero 的模板。 **提示**在训练过程中将被替换为具体的推理问题。

2.2.2. 奖励建模

奖励是训练信号的来源，它决定了强化学习的优化方向。为了训练 DeepSeek-R1-Zero，我们采用基于规则的奖励系统，主要由两种类型的奖励组成：

- **准确率奖励:** 准确率奖励模型评估回答是否正确。例如，在具有确定性结果的数学问题中，模型需要以指定格式（例如，在方框内）提供最终答案，以便进行可靠的基于规则的正确性验证。类似地，对于 LeetCode 问题，可以使用编译器根据预定义的测试用例生成反馈。
- **格式奖励:** 除了准确性奖励模型，我们还采用格式奖励模型，该模型强制模型将其思考过程放在 ‘<think>’ 和 ‘</think>’ 标签之间。

在开发 DeepSeek-R1-Zero 时，我们不应用结果或过程神经奖励模型，因为我们发现神经奖励模型在大规模强化学习过程中可能会受到奖励攻击，并且重新训练奖励模型需要额外的训练资源，这会使整个训练流程变得复杂。

2.2.3. 训练模板

为了训练 DeepSeek-R1-Zero，我们首先设计了一个简单的模板，该模板指导基础模型遵循我们指定的指令。如表1所示，此模板要求 DeepSeek-R1-Zero 首先生成推理过程，然后给出最终答案。我们有意将约束限制在这个结构格式中，避免任何特定内容偏差——例如强制反思性推理或推广特定的问题解决策略——以确保我们能够在RL过程中准确观察模型的自然进展。

2.2.4. 性能、自我进化过程和**DeepSeek-R1-Zero**的顿悟时刻

DeepSeek-R1-Zero的性能 图2描绘了DeepSeek-R1-Zero在AIME 2024基准测试中整个RL训练过程中的性能轨迹。如图所示，随着RL训练的推进，DeepSeek-R1-Zero的性能表现出稳定且持续提升。值得注意的是，AIME 2024的平均pass@1分数显著提高，从最初的15.6%跃升至令人印象深刻的71.0%，达到了与OpenAI-o1-0912相当的性能水平。这一显著改进突显了我们的RL算法在随时间优化模型性能方面的有效性。

表2提供了DeepSeek-R1-Zero和OpenAI的o1-0912模型在多种推理相关基准测试中的对比分析。研究结果表明，RL使

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

Table 2 | Comparison of DeepSeek-R1-Zero and OpenAI o1 models on reasoning-related benchmarks.

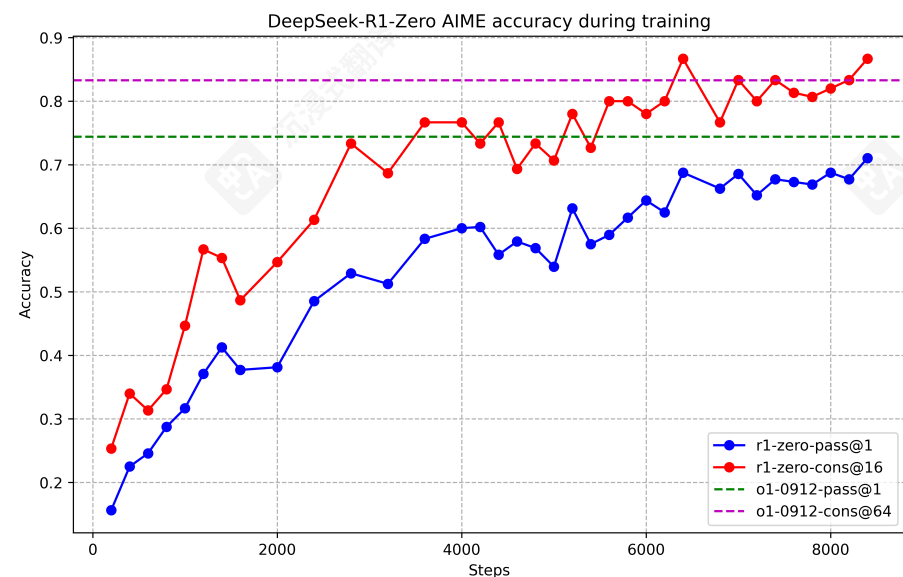


Figure 2 | AIME accuracy of DeepSeek-R1-Zero during training. For each question, we sample 16 responses and calculate the overall average accuracy to ensure a stable evaluation.

DeepSeek-R1-Zero to attain robust reasoning capabilities without the need for any supervised fine-tuning data. This is a noteworthy achievement, as it underscores the model’s ability to learn and generalize effectively through RL alone. Additionally, the performance of DeepSeek-R1-Zero can be further augmented through the application of majority voting. For example, when majority voting is employed on the AIME benchmark, DeepSeek-R1-Zero’s performance escalates from 71.0% to 86.7%, thereby exceeding the performance of OpenAI-o1-0912. The ability of DeepSeek-R1-Zero to achieve such competitive performance, both with and without majority voting, highlights its strong foundational capabilities and its potential for further advancements in reasoning tasks.

Self-evolution Process of DeepSeek-R1-Zero The self-evolution process of DeepSeek-R1-Zero is a fascinating demonstration of how RL can drive a model to improve its reasoning capabilities autonomously. By initiating RL directly from the base model, we can closely monitor the model’s progression without the influence of the supervised fine-tuning stage. This approach provides a clear view of how the model evolves over time, particularly in terms of its ability to handle complex reasoning tasks.

As depicted in Figure 3, the thinking time of DeepSeek-R1-Zero shows consistent improve-

模型	AIME 2024		MATH-500	GPQA 钻石	LiveCode 凳子	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	评分
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
OpenAI-o1-0912	74.4	83.3	94.8	77.3	63.4	1843
DeepSeek-R1-Zero	71.0	86.7	95.9	73.3	50.0	1444

表 2 | DeepSeek-R1-Zero 和 OpenAI o1 模型在推理相关基准测试上的比较。

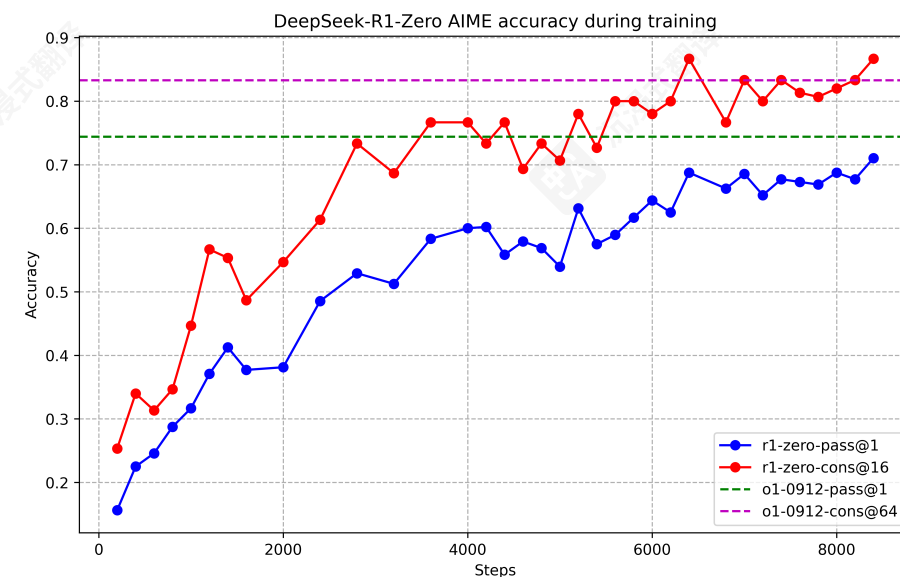


图 2 | DeepSeek-R1-Zero 在训练期间的 AIME 准确率。对于每个问题，我们采样 16 个响应并计算整体平均准确率以确保稳定的评估。

DeepSeek-R1-Zero 无需任何监督微调数据即可获得强大的推理能力。这是一个值得注意的成就，因为它突显了该模型仅通过强化学习（RL）就能有效学习和泛化的能力。此外，通过应用多数投票，DeepSeek-R1-Zero 的性能可以得到进一步提升。例如，在 AIME 基准测试中应用多数投票时，DeepSeek-R1-Zero 的性能从 71.0% 提升至 86.7%，从而超过了 OpenAI-o1-0912 的性能。DeepSeek-R1-Zero 无论是使用还是不使用多数投票都能取得如此具有竞争力的性能，这突显了其强大的基础能力，并表明其在推理任务方面具有进一步发展的潜力。

DeepSeek-R1-Zero 的自进化过程 DeepSeek-R1-Zero 的自进化过程是一个引人入胜的演示，展示了强化学习（RL）如何驱动模型自主提升其推理能力。通过直接从基础模型启动强化学习，我们可以在不受监督微调阶段影响的情况下密切监控模型的进展。这种方法清晰地展示了模型随时间的变化，特别是在处理复杂推理任务方面的能力。

如图3所示，DeepSeek-R1-Zero的思考时间显示出持续改进

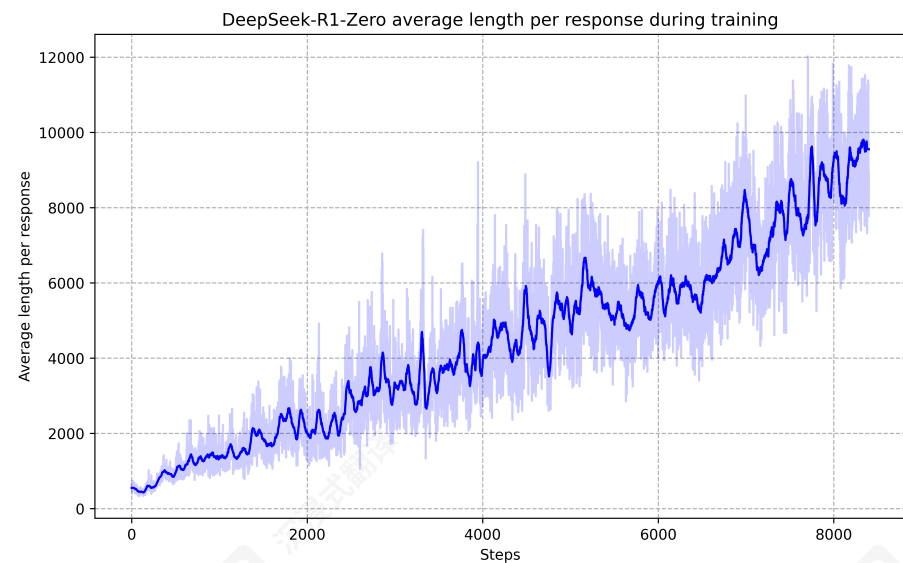


Figure 3 | The average response length of DeepSeek-R1-Zero on the training set during the RL process. DeepSeek-R1-Zero naturally learns to solve reasoning tasks with more thinking time.

ment throughout the training process. This improvement is not the result of external adjustments but rather an intrinsic development within the model. DeepSeek-R1-Zero naturally acquires the ability to solve increasingly complex reasoning tasks by leveraging extended test-time computation. This computation ranges from generating hundreds to thousands of reasoning tokens, allowing the model to explore and refine its thought processes in greater depth.

One of the most remarkable aspects of this self-evolution is the emergence of sophisticated behaviors as the test-time computation increases. Behaviors such as reflection—where the model revisits and reevaluates its previous steps—and the exploration of alternative approaches to problem-solving arise spontaneously. These behaviors are not explicitly programmed but instead emerge as a result of the model’s interaction with the reinforcement learning environment. This spontaneous development significantly enhances DeepSeek-R1-Zero’s reasoning capabilities, enabling it to tackle more challenging tasks with greater efficiency and accuracy.

Aha Moment of DeepSeek-R1-Zero A particularly intriguing phenomenon observed during the training of DeepSeek-R1-Zero is the occurrence of an “aha moment”. This moment, as illustrated in Table 3, occurs in an intermediate version of the model. During this phase, DeepSeek-R1-Zero learns to allocate more thinking time to a problem by reevaluating its initial approach. This behavior is not only a testament to the model’s growing reasoning abilities but also a captivating example of how reinforcement learning can lead to unexpected and sophisticated outcomes.

This moment is not only an “aha moment” for the model but also for the researchers observing its behavior. It underscores the power and beauty of reinforcement learning: rather than explicitly teaching the model on how to solve a problem, we simply provide it with the right incentives, and it autonomously develops advanced problem-solving strategies. The “aha moment” serves as a powerful reminder of the potential of RL to unlock new levels of intelligence in artificial systems, paving the way for more autonomous and adaptive models in the future.

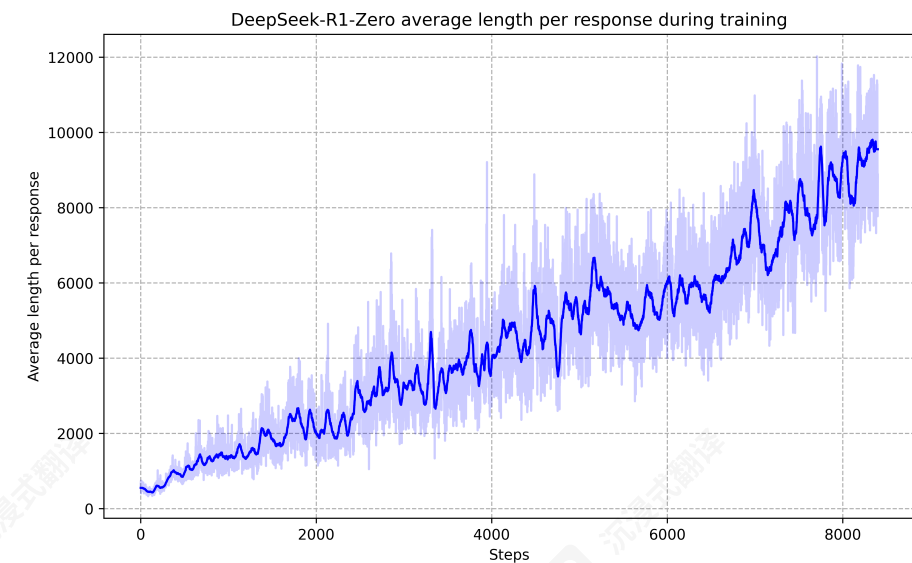


图 3 | 在RL过程中DeepSeek-R1-Zero在训练集上的平均响应长度。DeepSeek-R1-Zero自然地学会用更多思考时间来解决问题任务。

在整个训练过程中。这种改进不是外部调整的结果，而是模型内部的内在发展。

DeepSeek-R1-Zero通过利用扩展的测试时间计算，自然地获得了解决越来越复杂的推理任务的能力。这种计算范围从生成数百到数千个推理标记，允许模型更深入地探索和改进其思维过程。

这种自我进化的最显著方面之一是随着测试时间计算的增加而出现的复杂行为。例如反思——模型回顾和重新评估其先前步骤的行为——以及探索解决问题替代方法的行为自发出现。这些行为没有被显式编程，而是作为模型与强化学习环境交互的结果而出现的。这种自发发展显著增强了DeepSeek-R1-Zero的推理能力，使其能够更高效、更准确地处理更具挑战性的任务。

DeepSeek-R1-Zero 的“啊哈时刻” 在 DeepSeek-R1-Zero 的训练过程中观察到的一个特别引人入胜的现象是“啊哈时刻”的发生。如表 3 所示，这个时刻发生在模型的某个中间版本。在这个阶段，DeepSeek-R1-Zero 通过重新评估其初始方法来学习为问题分配更多思考时间。这种行为不仅证明了模型推理能力的增长，而且也是强化学习如何导致意想不到的复杂结果的迷人例证。

这一刻不仅对模型而言是一个“啊哈时刻”，对观察其行为的科研人员来说也是。它突显了强化学习的力量 and 美妙：我们并非明确教导模型如何解决问题，而是仅仅为其提供正确的激励，它便能自主地发展出高级的问题解决策略。“啊哈时刻”作为一个强有力的提醒，展现了强化学习解锁人工智能系统新智能水平的潜力，为未来更自主和适应性更强的模型铺平了道路。

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$(\sqrt{a - \sqrt{a + x}})^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both ...

$$(\sqrt{a - \sqrt{a + x}})^2 = x^2 \implies a - \sqrt{a + x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a + x}} = x$$

First, let's square both sides:

$$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

Drawback of DeepSeek-R1-Zero Although DeepSeek-R1-Zero exhibits strong reasoning capabilities and autonomously develops unexpected and powerful reasoning behaviors, it faces several issues. For instance, DeepSeek-R1-Zero struggles with challenges like poor readability, and language mixing. To make reasoning processes more readable and share them with the open community, we explore DeepSeek-R1, a method that utilizes RL with human-friendly cold-start data.

DeepSeek-R1-Zero 的缺点 尽管 DeepSeek-R1-Zero 表现出强大的推理能力，并能自主开发意外且强大的推理行为，但它仍面临一些问题。例如，DeepSeek-R1-Zero 在处理可读性差、语言混杂等挑战时存在困难。为了使推理过程更易于理解并与开放社区分享，我们探索了 DeepSeek-R1，这是一种利用带有人类友好型冷启动数据的强化学习的方法。

2.3. DeepSeek-R1: Reinforcement Learning with Cold Start

Inspired by the promising results of DeepSeek-R1-Zero, two natural questions arise: 1) Can reasoning performance be further improved or convergence accelerated by incorporating a small amount of high-quality data as a cold start? 2) How can we train a user-friendly model that not only produces clear and coherent Chains of Thought (CoT) but also demonstrates strong general capabilities? To address these questions, we design a pipeline to train DeepSeek-R1. The pipeline consists of four stages, outlined as follows.

2.3. DeepSeek-R1: 带有冷启动的强化学习

受 DeepSeek-R1-Zero 令人鼓舞的成果启发，产生了两个自然的问题：1) 是否可以通过引入少量高质量数据作为冷启动来进一步提高推理性能或加速收敛？2) 我们如何训练一个用户友好的模型，该模型不仅产生清晰连贯的思维链 (CoT)，而且展现出强大的泛化能力？为了解决这些问题，我们设计了一个用于训练 DeepSeek-R1 的流程。该流程由四个阶段组成，概述如下。

2.3.1. Cold Start

Unlike DeepSeek-R1-Zero, to prevent the early unstable cold start phase of RL training from the base model, for DeepSeek-R1 we construct and collect a small amount of long CoT data to fine-tune the model as the initial RL actor. To collect such data, we have explored several approaches: using few-shot prompting with a long CoT as an example, directly prompting models to generate detailed answers with reflection and verification, gathering DeepSeek-R1-Zero outputs in a readable format, and refining the results through post-processing by human annotators.

2.3.1. 冷启动

与 DeepSeek-R1-Zero 不同，为了防止基础模型中 RL 训练的早期不稳定冷启动阶段，对于 DeepSeek-R1，我们构建并收集少量长 CoT 数据，将其作为初始 RL 行为者来微调模型。为了收集此类数据，我们探索了多种方法：使用包含长 CoT 的少样本提示，直接提示模型生成带有反思和验证的详细答案，收集 DeepSeek-R1-Zero 的可读格式输出，以及通过人工标注者的后处理来优化结果。

In this work, we collect thousands of cold-start data to fine-tune the DeepSeek-V3-Base as the starting point for RL. Compared to DeepSeek-R1-Zero, the advantages of cold start data

在这项工作中，我们收集数千个冷启动数据来微调 DeepSeek-V3-Base 作为 RL 的起点。与 DeepSeek-R1-Zero 相比，冷启动数据的优势

include:

- **Readability:** A key limitation of DeepSeek-R1-Zero is that its content is often not suitable for reading. Responses may mix multiple languages or lack markdown formatting to highlight answers for users. In contrast, when creating cold-start data for DeepSeek-R1, we design a readable pattern that includes a summary at the end of each response and filters out responses that are not reader-friendly. Here, we define the output format as `|special_token|<reasoning_process>|special_token|<summary>`, where the reasoning process is the CoT for the query, and the summary is used to summarize the reasoning results.
- **Potential:** By carefully designing the pattern for cold-start data with human priors, we observe better performance against DeepSeek-R1-Zero. We believe the iterative training is a better way for reasoning models.

2.3.2. Reasoning-oriented Reinforcement Learning

After fine-tuning DeepSeek-V3-Base on the cold start data, we apply the same large-scale reinforcement learning training process as employed in DeepSeek-R1-Zero. This phase focuses on enhancing the model's reasoning capabilities, particularly in reasoning-intensive tasks such as coding, mathematics, science, and logic reasoning, which involve well-defined problems with clear solutions. During the training process, we observe that CoT often exhibits language mixing, particularly when RL prompts involve multiple languages. To mitigate the issue of language mixing, we introduce a language consistency reward during RL training, which is calculated as the proportion of target language words in the CoT. Although ablation experiments show that such alignment results in a slight degradation in the model's performance, this reward aligns with human preferences, making it more readable. Finally, we combine the accuracy of reasoning tasks and the reward for language consistency by directly summing them to form the final reward. We then apply RL training on the fine-tuned model until it achieves convergence on reasoning tasks.

2.3.3. Rejection Sampling and Supervised Fine-Tuning

When reasoning-oriented RL converges, we utilize the resulting checkpoint to collect SFT (Supervised Fine-Tuning) data for the subsequent round. Unlike the initial cold-start data, which primarily focuses on reasoning, this stage incorporates data from other domains to enhance the model's capabilities in writing, role-playing, and other general-purpose tasks. Specifically, we generate the data and fine-tune the model as described below.

Reasoning data We curate reasoning prompts and generate reasoning trajectories by performing rejection sampling from the checkpoint from the above RL training. In the previous stage, we only included data that could be evaluated using rule-based rewards. However, in this stage, we expand the dataset by incorporating additional data, some of which use a generative reward model by feeding the ground-truth and model predictions into DeepSeek-V3 for judgment. Additionally, because the model output is sometimes chaotic and difficult to read, we have filtered out chain-of-thought with mixed languages, long paragraphs, and code blocks. For each prompt, we sample multiple responses and retain only the correct ones. In total, we collect about 600k reasoning related training samples.

include:

- **可读性:** DeepSeek-R1-Zero的一个关键限制是其内容通常不适合阅读。响应可能混合多种语言或缺乏markdown格式来突出显示答案以供用户使用。相比之下,在为DeepSeek-R1创建冷启动数据时,我们设计了一种可读模式,其中包括在每个响应的末尾添加摘要,并过滤掉不便于阅读的响应。在这里,我们将输出格式定义为|特殊标记|<推理过程>|特殊标记|<摘要>,其中推理过程是查询的CoT,摘要用于总结推理结果。
- **潜力:** 通过精心设计包含人类先验的冷启动数据模式,我们观察到在DeepSeek-R1-Zero上的性能更好。我们相信迭代训练是推理模型的更好方式。

2.3.2. 推理导向的强化学习

在冷启动数据上微调DeepSeek-V3-Base后,我们应用了与DeepSeek-R1-Zero相同的规模化强化学习训练过程。这一阶段专注于增强模型的推理能力,特别是在编码、数学、科学和逻辑推理等推理密集型任务中,这些任务涉及定义明确的问题和明确的解决方案。在训练过程中,我们观察到CoT经常出现语言混合现象,尤其是在RL提示涉及多种语言时。为了缓解语言混合问题,我们在RL训练中引入了语言一致性奖励,该奖励计算为目标语言单词在CoT中的比例。尽管消融实验表明这种对齐会导致模型性能略有下降,但该奖励符合人类偏好,使其更易于阅读。最后,我们将推理任务的准确性和语言一致性奖励直接相加,形成最终奖励。然后,我们对微调后的模型应用RL训练,直到它在推理任务上达到收敛。

2.3.3. 拒绝采样与监督微调

当面向推理的强化学习收敛时,我们利用生成的检查点来收集下一轮的监督微调(SFT)数据。与最初侧重于推理的冷启动数据不同,这一阶段结合了其他领域的数据,以增强模型在写作、角色扮演和其他通用任务中的能力。具体来说,我们按照以下方式生成数据并微调模型。

推理数据 我们通过从上述强化学习训练生成的检查点进行拒绝采样来整理推理提示并生成推理轨迹。在上一阶段,我们仅包含可以使用基于规则的奖励进行评估的数据。然而,在这一阶段,我们通过结合额外数据来扩展数据集,其中一些数据通过将真实值和模型预测输入到DeepSeek-V3进行判断,使用生成式奖励模型。此外,由于模型输出有时混乱且难以阅读,我们已经过滤出混合语言、长段落和代码块等思维链。对于每个提示,我们采样多个响应并仅保留正确的响应。总共,我们收集了约600k个与推理相关的训练样本。

Non-Reasoning data For non-reasoning data, such as writing, factual QA, self-cognition, and translation, we adopt the DeepSeek-V3 pipeline and reuse portions of the SFT dataset of DeepSeek-V3. For certain non-reasoning tasks, we call DeepSeek-V3 to generate a potential chain-of-thought before answering the question by prompting. However, for simpler queries, such as “hello” we do not provide a CoT in response. In the end, we collected a total of approximately 200k training samples that are unrelated to reasoning.

We fine-tune DeepSeek-V3-Base for two epochs using the above curated dataset of about 800k samples.

2.3.4. Reinforcement Learning for all Scenarios

To further align the model with human preferences, we implement a secondary reinforcement learning stage aimed at improving the model’s helpfulness and harmlessness while simultaneously refining its reasoning capabilities. Specifically, we train the model using a combination of reward signals and diverse prompt distributions. For reasoning data, we adhere to the methodology outlined in DeepSeek-R1-Zero, which utilizes rule-based rewards to guide the learning process in math, code, and logical reasoning domains. For general data, we resort to reward models to capture human preferences in complex and nuanced scenarios. We build upon the DeepSeek-V3 pipeline and adopt a similar distribution of preference pairs and training prompts. For helpfulness, we focus exclusively on the final summary, ensuring that the assessment emphasizes the utility and relevance of the response to the user while minimizing interference with the underlying reasoning process. For harmlessness, we evaluate the entire response of the model, including both the reasoning process and the summary, to identify and mitigate any potential risks, biases, or harmful content that may arise during the generation process. Ultimately, the integration of reward signals and diverse data distributions enables us to train a model that excels in reasoning while prioritizing helpfulness and harmlessness.

2.4. Distillation: Empower Small Models with Reasoning Capability

To equip more efficient smaller models with reasoning capabilities like DeepSeek-R1, we directly fine-tuned open-source models like Qwen (Qwen, 2024b) and Llama (AI@Meta, 2024) using the 800k samples curated with DeepSeek-R1, as detailed in §2.3.3. Our findings indicate that this straightforward distillation method significantly enhances the reasoning abilities of smaller models. The base models we use here are Qwen2.5-Math-1.5B, Qwen2.5-Math-7B, Qwen2.5-14B, Qwen2.5-32B, Llama-3.1-8B, and Llama-3.3-70B-Instruct. We select Llama-3.3 because its reasoning capability is slightly better than that of Llama-3.1.

For distilled models, we apply only SFT and do not include an RL stage, even though incorporating RL could substantially boost model performance. Our primary goal here is to demonstrate the effectiveness of the distillation technique, leaving the exploration of the RL stage to the broader research community.

3. Experiment

Benchmarks We evaluate models on MMLU (Hendrycks et al., 2020), MMLU-Redux (Gema et al., 2024), MMLU-Pro (Wang et al., 2024), C-Eval (Huang et al., 2023), and CMMLU (Li et al., 2023), IFEval (Zhou et al., 2023), FRAMES (Krishna et al., 2024), GPQA Diamond (Rein et al., 2023), SimpleQA (OpenAI, 2024c), C-SimpleQA (He et al., 2024), SWE-Bench Verified (OpenAI,

非推理数据 对于非推理数据，例如写作、事实性问答、自我认知和翻译，我们采用 DeepSeek-V3 管道并重用 DeepSeek-V3 的 SFT 数据集的部分内容。对于某些非推理任务，我们通过提示 DeepSeek-V3 在回答问题之前生成一个潜在的思维链。然而，对于更简单的查询，例如“你好”，我们不会提供思维链作为回应。最后，我们收集了大约 200k 个与推理无关的训练样本。

我们使用上述精选的约 800k 个样本数据集对 DeepSeek-V3-Base 进行两个 epoch 的微调。

2.3.4. 所有场景的强化学习

为了进一步使模型与人类偏好保持一致，我们实施了一个次要的强化学习阶段，旨在提高模型的有用性和无害性，同时改进其推理能力。具体来说，我们使用奖励信号和多样化的提示分布的组合来训练模型。对于推理数据，我们遵循 DeepSeek-R1-Zero 中概述的方法论，该方法利用基于规则的奖励来指导数学、代码和逻辑推理领域的学习过程。对于一般数据，我们采用奖励模型来捕捉复杂和微妙场景中的人类偏好。我们基于 DeepSeek-V3 管道，并采用类似的偏好对和训练提示的分布。对于有用性，我们专注于最终摘要，确保评估强调响应用户的效用和相关性，同时尽量减少对底层推理过程的干扰。对于无害性，我们评估模型的整体响应，包括推理过程和摘要，以识别和减轻在生成过程中可能出现的任何潜在风险、偏见或有害内容。最终，奖励信号和多样化数据分布的集成使我们能够训练一个在推理方面表现出色，同时优先考虑有用性和无害性的模型。

2.4. 蒸馏：赋予Small模型推理能力

为了给更高效的较小模型赋予推理能力（如DeepSeek-R1），我们直接使用DeepSeek-R1精心策划的800k样本对开源模型（如Qwen（Qwen, 2024b）和Llama（AI@Meta, 2024））进行微调，具体细节见 § 2.3.3。我们的研究表明，这种直接的蒸馏方法显著提升了较小模型的推理能力。我们在此使用的基模型包括Qwen2.5-Math-1.5B、Qwen2.5-Math-7B、Qwen2.5-14B、Qwen2.5-32B、Llama-3.1-8B和Llama-3.3-70B-Instruct。我们选择Llama-3.3是因为其推理能力略优于Llama-3.1。

对于蒸馏后的模型，我们仅应用SFT，不包含RL阶段，尽管加入RL可以显著提升模型性能。我们的主要目标在这里是展示蒸馏技术的有效性，将RL阶段的探索留给更广泛的研究社区。

3. 实验

基准测试 We evaluate models on MMLU (Hendrycks et al., 2020), MMLU-Redux (Gema et al., 2024), MMLU-Pro (Wang et al., 2024), C-Eval (Huang et al., 2023), and CMMLU (Li et al., 2023), IFEval (Zhou et al., 2023), FRAMES (Krishna et al., 2024), GPQA Diamond (Rein et al., 2023), SimpleQA (OpenAI, 2024c), C-SimpleQA (He et al., 2024), SWE-Bench Verified (OpenAI,

2024d), Aider ¹, LiveCodeBench (Jain et al., 2024) (2024-08 – 2025-01), Codeforces ², Chinese National High School Mathematics Olympiad (CNMO 2024)³, and American Invitational Mathematics Examination 2024 (AIME 2024) (MAA, 2024). In addition to standard benchmarks, we also evaluate our models on open-ended generation tasks using LLMs as judges. Specifically, we adhere to the original configurations of AlpacaEval 2.0 (Dubois et al., 2024) and Arena-Hard (Li et al., 2024), which leverage GPT-4-Turbo-1106 as judges for pairwise comparisons. Here, we only feed the final summary to evaluation to avoid the length bias. For distilled models, we report representative results on AIME 2024, MATH-500, GPQA Diamond, Codeforces, and LiveCodeBench.

Evaluation Prompts Following the setup in DeepSeek-V3, standard benchmarks such as MMLU, DROP, GPQA Diamond, and SimpleQA are evaluated using prompts from the simple-evals framework. For MMLU-Redux, we adopt the Zero-Eval prompt format (Lin, 2024) in a zero-shot setting. In terms of MMLU-Pro, C-Eval and CLUE-WSC, since the original prompts are few-shot, we slightly modify the prompt to the zero-shot setting. The CoT in few-shot may hurt the performance of DeepSeek-R1. Other datasets follow their original evaluation protocols with default prompts provided by their creators. For code and math benchmarks, the HumanEval-Mul dataset covers eight mainstream programming languages (Python, Java, C++, C#, JavaScript, TypeScript, PHP, and Bash). Model performance on LiveCodeBench is evaluated using CoT format, with data collected between August 2024 and January 2025. The Codeforces dataset is evaluated using problems from 10 Div.2 contests along with expert-crafted test cases, after which the expected ratings and percentages of competitors are calculated. SWE-Bench verified results are obtained via the agentless framework (Xia et al., 2024). AIDER-related benchmarks are measured using a "diff" format. DeepSeek-R1 outputs are capped at a maximum of 32,768 tokens for each benchmark.

Baselines We conduct comprehensive evaluations against several strong baselines, including DeepSeek-V3, Claude-Sonnet-3.5-1022, GPT-4o-0513, OpenAI-o1-mini, and OpenAI-o1-1217. Since accessing the OpenAI-o1-1217 API is challenging in mainland China, we report its performance based on official reports. For distilled models, we also compare the open-source model QwQ-32B-Preview (Qwen, 2024a).

Evaluation Setup We set the maximum generation length to 32,768 tokens for the models. We found that using greedy decoding to evaluate long-output reasoning models results in higher repetition rates and significant variability across different checkpoints. Therefore, we default to pass@k evaluation (Chen et al., 2021) and report pass@1 using a non-zero temperature. Specifically, we use a sampling temperature of 0.6 and a top-p value of 0.95 to generate k responses (typically between 4 and 64, depending on the test set size) for each question. Pass@1 is then calculated as

$$\text{pass@1} = \frac{1}{k} \sum_{i=1}^k p_i,$$

where p_i denotes the correctness of the i -th response. This method provides more reliable performance estimates. For AIME 2024, we also report consensus (majority vote) results (Wang et al., 2022) using 64 samples, denoted as cons@64.

¹<https://aider.chat>

²<https://codeforces.com>

³<https://www.cms.org.cn/Home/comp/comp/cid/12.html>

2024d), Aider 1, LiveCodeBench (Jain 等人, 2024) (2024-08 – 2025-01), Codeforces 2, 中国全国高中数学奥林匹克竞赛 (CNMO 2024)³, 和美国邀请数学考试 2024 (AIME 2024) (MAA, 2024)。除了标准基准测试外, 我们还使用 LLM 作为裁判对开放生成任务进行评估。具体来说, 我们遵循 AlpacaEval 2.0 (Dubois 等人, 2024) 和 Arena-Hard (Li 等人, 2024) 的原始配置, 后者使用 GPT-4-Turbo-1106 作为成对比较的裁判。在这里, 我们仅将最终摘要提供给评估以避免长度偏差。对于蒸馏模型, 我们在 AIME 2024、MATH-500、GPQA Diamond、Codeforces 和 LiveCodeBench 上报告了代表性结果。

评估提示 遵循 DeepSeek-V3 的设置, 使用 simple-evals 框架中的提示对 MMLU、DROP、GPQA Diamond 和 SimpleQA 等标准基准进行评估。对于 MMLU-Redux, 我们在零样本设置中采用 Zero-Eval 提示格式 (Lin, 2024)。在 MMLU-Pro、C-Eval 和 CLUE-WSC 方面, 由于原始提示是少样本的, 我们将提示稍微修改为零样本设置。少样本中的 CoT 可能会损害 DeepSeek-R1 的性能。其他数据集遵循其原始评估协议, 并使用其创建者提供的默认提示。对于代码和数学基准, HumanEval-Mul 数据集涵盖了八种主流编程语言 (Python、Java、C++、C#、JavaScript、TypeScript、PHP 和 Bash)。模型在 LiveCodeBench 上的性能使用 CoT 格式进行评估, 数据收集时间在 2024 年 8 月至 2025 年 1 月之间。Codeforces 数据集使用 10 场 Div.2 比赛的问题以及专家设计的测试用例进行评估, 之后计算竞争对手的预期评分和百分比。SWE-Bench 验证结果通过无代理框架 (Xia 等人, 2024) 获得。AIDER 相关基准使用 "diff" 格式进行测量。DeepSeek-R1 的输出对每个基准最多限制为 32,768 个 token。

基线 我们对多个强基线进行了全面评估, 包括 DeepSeek-V3、Claude-Sonnet-3.5-1022、GPT-4o-0513、OpenAI-o1-mini 和 OpenAI-o1-1217。由于在中国大陆访问 OpenAI-o1-1217 API 存在挑战, 我们根据官方报告报告其性能。对于蒸馏模型, 我们也比较了开源模型 QwQ-32B-Preview (Qwen, 2024a)。

评估设置 我们将模型的最大生成长度设置为 32,768 个 token。我们发现使用贪婪解码来评估长输出推理模型会导致更高的重复率, 并且在不同的检查点之间存在显著差异。因此, 我们默认使用 pass@k 评估 (Chen 等人, 2021 年) 并使用非零温度报告 pass@1。具体来说, 我们使用 0.6 的采样温度和 0.95 的 top-p 值来为每个问题生成 k 响应 (通常在 4 到 64 之间, 具体取决于测试集大小)。然后计算 pass@1 为

$$\text{pass@1} = \frac{1}{k} \sum_{i=1}^k p_i,$$

where p_i denotes the correctness of the i -th response. This method provides more reliable performance estimates. For AIME 2024, we also report consensus (majority vote) results (Wang et al., 2022) using 64 samples, denoted as cons@64.

¹<https://aider.chat>

²<https://codeforces.com>

³<https://www.cms.org.cn/Home/comp/comp/cid/12.html>

3.1. DeepSeek-R1 Evaluation

Benchmark (Metric)		Claude-3.5- Sonnet-1022	GPT-4o 0513	DeepSeek V3	OpenAI o1-mini	OpenAI o1-1217	DeepSeek R1
	Architecture	-	-	MoE	-	-	MoE
	# Activated Params	-	-	37B	-	-	37B
	# Total Params	-	-	671B	-	-	671B
English	MMLU (Pass@1)	88.3	87.2	88.5	85.2	91.8	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	92.9
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	84.0
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	92.2
	IF-Eval (Prompt Strict)	86.5	84.3	86.1	84.8	-	83.3
	GPQA Diamond (Pass@1)	65.0	49.9	59.1	60.0	75.7	71.5
	SimpleQA (Correct)	28.4	38.2	24.9	7.0	47.0	30.1
	FRAMES (Acc.)	72.5	80.5	73.3	76.9	-	82.5
	AlpacaEval2.0 (LC-winrate)	52.0	51.1	70.0	57.8	-	87.6
	ArenaHard (GPT-4-1106)	85.2	80.4	85.5	92.0	-	92.3
Code	LiveCodeBench (Pass@1-COT)	38.9	32.9	36.2	53.8	63.4	65.9
	Codeforces (Percentile)	20.3	23.6	58.7	93.4	96.6	96.3
	Codeforces (Rating)	717	759	1134	1820	2061	2029
	SWE Verified (Resolved)	50.8	38.8	42.0	41.6	48.9	49.2
	Aider-Polyglot (Acc.)	45.3	16.0	49.6	32.9	61.7	53.3
Math	AIME 2024 (Pass@1)	16.0	9.3	39.2	63.6	79.2	79.8
	MATH-500 (Pass@1)	78.3	74.6	90.2	90.0	96.4	97.3
	CNMO 2024 (Pass@1)	13.1	10.8	43.2	67.6	-	78.8
Chinese	CLUEWSC (EM)	85.4	87.9	90.9	89.9	-	92.8
	C-Eval (EM)	76.7	76.0	86.5	68.9	-	91.8
	C-SimpleQA (Correct)	55.4	58.7	68.0	40.3	-	63.7

Table 4 | Comparison between DeepSeek-R1 and other representative models.

For education-oriented knowledge benchmarks such as MMLU, MMLU-Pro, and GPQA Diamond, DeepSeek-R1 demonstrates superior performance compared to DeepSeek-V3. This improvement is primarily attributed to enhanced accuracy in STEM-related questions, where significant gains are achieved through large-scale reinforcement learning. Additionally, DeepSeek-R1 excels on FRAMES, a long-context-dependent QA task, showcasing its strong document analysis capabilities. This highlights the potential of reasoning models in AI-driven search and data analysis tasks. On the factual benchmark SimpleQA, DeepSeek-R1 outperforms DeepSeek-V3, demonstrating its capability in handling fact-based queries. A similar trend is observed where OpenAI-o1 surpasses GPT-4o on this benchmark. However, DeepSeek-R1 performs worse than DeepSeek-V3 on the Chinese SimpleQA benchmark, primarily due to its tendency to refuse answering certain queries after safety RL. Without safety RL, DeepSeek-R1 could achieve an accuracy of over 70%.

DeepSeek-R1 also delivers impressive results on IF-Eval, a benchmark designed to assess a model’s ability to follow format instructions. These improvements can be linked to the inclusion of instruction-following data during the final stages of supervised fine-tuning (SFT) and RL training. Furthermore, remarkable performance is observed on AlpacaEval2.0 and ArenaHard, indicating DeepSeek-R1’s strengths in writing tasks and open-domain question answering. Its significant outperformance of DeepSeek-V3 underscores the generalization benefits of large-scale RL, which not only boosts reasoning capabilities but also improves performance across diverse domains. Moreover, the summary lengths generated by DeepSeek-R1 are concise, with an average of 689 tokens on ArenaHard and 2,218 characters on AlpacaEval 2.0. This indicates that

3.1. DeepSeek-R1 Evaluation

基准 (Metric)		Claude-3.5- Sonnet-1022	GPT-4o 0513	DeepSeek V3	OpenAI o1-mini	OpenAI o1-1217	DeepSeek R1
English	Architecture	-	-	MoE	-	-	MoE
	# Activated Params	-	-	37B	-	-	37B
	# 总参数	-	-	671B	-	-	671B
	MMLU (Pass@1)	88.3	87.2	88.5	85.2	91.8	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	92.9
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	84.0
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	92.2
	IF-Eval (Prompt Strict)	86.5	84.3	86.1	84.8	-	83.3
	GPQA Diamond (Pass@1)	65.0	49.9	59.1	60.0	75.7	71.5
	SimpleQA (Correct)	28.4	38.2	24.9	7.0	47.0	30.1
Code	FRAMES (Acc.)	72.5	80.5	73.3	76.9	-	82.5
	AlpacaEval2.0 (LC-winrate)	52.0	51.1	70.0	57.8	-	87.6
	ArenaHard (GPT-4-1106)	85.2	80.4	85.5	92.0	-	92.3
	LiveCodeBench (Pass@1-COT)	38.9	32.9	36.2	53.8	63.4	65.9
	Codeforces (Percentile)	20.3	23.6	58.7	93.4	96.6	96.3
Math	Codeforces (Rating)	717	759	1134	1820	2061	2029
	SWE Verified (Resolved)	50.8	38.8	42.0	41.6	48.9	49.2
	Aider-Polyglot (Acc.)	45.3	16.0	49.6	32.9	61.7	53.3
中文	AIME 2024 (Pass@1)	16.0	9.3	39.2	63.6	79.2	79.8
	MATH-500 (Pass@1)	78.3	74.6	90.2	90.0	96.4	97.3
	CNMO 2024 (Pass@1)	13.1	10.8	43.2	67.6	-	78.8
中文	CLUEWSC (EM)	85.4	87.9	90.9	89.9	-	92.8
	C-Eval (EM)	76.7	76.0	86.5	68.9	-	91.8
	C-SimpleQA (Correct)	55.4	58.7	68.0	40.3	-	63.7

表 4 | DeepSeek-R1 与其他代表性模型的对比。

在教育导向的知识基准测试（如 MMLU、MMLU-Pro 和 GPQA Diamond）中，DeepSeek-R1 的表现优于 DeepSeek-V3。这种改进主要归因于 STEM 相关问题上的准确率提升，通过大规模强化学习实现了显著增益。此外，DeepSeek-R1 在 FRAMES（一个长上下文依赖的问答任务）上表现出色，展示了其强大的文档分析能力。这突显了推理模型在 AI 驱动的搜索和数据分析任务中的潜力。在事实基准测试 SimpleQA 上，DeepSeek-R1 优于 DeepSeek-V3，展示了其处理基于事实查询的能力。在这个基准测试上，OpenAI-o1 也超过了 GPT-4o，呈现出类似的趋势。然而，在中文 SimpleQA 基准测试上，DeepSeek-R1 的表现不如 DeepSeek-V3，这主要由于其倾向于在安全 RL 后拒绝回答某些查询。如果没有安全 RL，DeepSeek-R1 的准确率可以达到 70% 以上。

DeepSeek-R1 在 IF-Eval 上也取得了令人印象深刻的结果，IF-Eval 是一个用于评估模型遵循格式指令能力的基准。这些改进可以归因于在监督微调（SFT）和 RL 训练的最终阶段包含了遵循指令的数据。此外，在 AlpacaEval2.0 和 ArenaHard 上观察到显著的性能表现，这表明 DeepSeek-R1 在写作任务和开放域问答方面的优势。其显著优于 DeepSeek-V3 体现了大规模 RL 的泛化收益，这不仅提升了推理能力，还改善了跨不同领域的性能。此外，DeepSeek-R1 生成的摘要长度简洁，在 ArenaHard 上的平均长度为 689 个 token，在 AlpacaEval 2.0 上的平均长度为 2,218 个字符。这表明

DeepSeek-R1 avoids introducing length bias during GPT-based evaluations, further solidifying its robustness across multiple tasks.

On math tasks, DeepSeek-R1 demonstrates performance on par with OpenAI-o1-1217, surpassing other models by a large margin. A similar trend is observed on coding algorithm tasks, such as LiveCodeBench and Codeforces, where reasoning-focused models dominate these benchmarks. On engineering-oriented coding tasks, OpenAI-o1-1217 outperforms DeepSeek-R1 on Aider but achieves comparable performance on SWE Verified. We believe the engineering performance of DeepSeek-R1 will improve in the next version, as the amount of related RL training data currently remains very limited.

3.2. Distilled Model Evaluation

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

Table 5 | Comparison of DeepSeek-R1 distilled models and other comparable models on reasoning-related benchmarks.

As shown in Table 5, simply distilling DeepSeek-R1’s outputs enables the efficient DeepSeek-R1-7B (i.e., DeepSeek-R1-Distill-Qwen-7B, abbreviated similarly below) to outperform non-reasoning models like GPT-4o-0513 across the board. DeepSeek-R1-14B surpasses QwQ-32B-Preview on all evaluation metrics, while DeepSeek-R1-32B and DeepSeek-R1-70B significantly exceed o1-mini on most benchmarks. These results demonstrate the strong potential of distillation. Additionally, we found that applying RL to these distilled models yields significant further gains. We believe this warrants further exploration and therefore present only the results of the simple SFT-distilled models here.

4. Discussion

4.1. Distillation v.s. Reinforcement Learning

In Section 3.2, we can see that by distilling DeepSeek-R1, the small model can achieve impressive results. However, there is still one question left: can the model achieve comparable performance through the large-scale RL training discussed in the paper without distillation?

To answer this question, we conduct large-scale RL training on Qwen-32B-Base using math, code, and STEM data, training for over 10K steps, resulting in DeepSeek-R1-Zero-Qwen-32B. The experimental results, shown in Table 6, demonstrate that the 32B base model, after large-scale

DeepSeek-R1 在基于 GPT 的评估中避免引入长度偏差，进一步巩固了其在多项任务中的鲁棒性。

在数学任务上，DeepSeek-R1 的表现与 OpenAI-o1-1217 相当，大幅超越了其他模型。在编码算法任务上，如 LiveCodeBench 和 Codeforces，观察到类似的趋势，以推理为重点的模型在这些基准测试中占据主导地位。在面向工程的编码任务上，OpenAI-o1-1217 在 Aider 上优于 DeepSeek-R1，但在 SWE Verified 上表现相当。我们认为 DeepSeek-R1 的工程性能将在下一个版本中有所提升，因为目前相关的强化学习训练数据量仍然非常有限。

3.2. 摘要模型评估

模型	AIME 2024		MATH-500	GPQA 钻石	LiveCode 凳子	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	评分
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	1820
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	86.7	94.5	65.2	57.5	1633

Table 5 | DeepSeek-R1蒸馏模型与其他可比模型在推理相关基准上的比较。

如表5所示，仅蒸馏DeepSeek-R1的输出即可使高效的DeepSeek-R1-7B（即 DeepSeek-R1-Distill-Qwen-7B，以下类似简称）在各个方面都优于非推理模型如 GPT-4o-0513。DeepSeek-R1-14B在所有评估指标上都超越了QwQ-32B-Preview，而 DeepSeek-R1-32B和DeepSeek-R1-70B在大多数基准上显著超过了o1-mini。这些结果表明了蒸馏的强大潜力。此外，我们发现将这些蒸馏模型应用强化学习（RL）可以带来显著的进一步增益。我们认为这值得进一步探索，因此在此仅呈现简单SFT蒸馏模型的结果。

4. 讨论

4.1. 蒸馏 v.s. 强化学习

在 3.2 节中，我们可以看到通过蒸馏 DeepSeek-R1，小模型可以取得令人印象深刻的结果。然而，仍然有一个问题尚未解决：模型能否通过论文中讨论的大规模 RL 训练（不使用蒸馏）实现相当的性能？

为了回答这个问题，我们使用数学、代码和 STEM 数据在 Qwen-32B-Base 上进行大规模 RL 训练，训练超过 10K 步，最终得到 DeepSeek-R1-Zero-Qwen-32B。实验结果如表 6 所示，表明 32B 基础模型在大规模训练后，

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCodeBench
	pass@1	cons@64	pass@1	pass@1	pass@1
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6	55.0	40.2
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2

Table 6 | Comparison of distilled and RL Models on Reasoning-Related Benchmarks.

RL training, achieves performance on par with QwQ-32B-Preview. However, DeepSeek-R1-Distill-Qwen-32B, which is distilled from DeepSeek-R1, performs significantly better than DeepSeek-R1-Zero-Qwen-32B across all benchmarks.

Therefore, we can draw two conclusions: First, distilling more powerful models into smaller ones yields excellent results, whereas smaller models relying on the large-scale RL mentioned in this paper require enormous computational power and may not even achieve the performance of distillation. Second, while distillation strategies are both economical and effective, advancing beyond the boundaries of intelligence may still require more powerful base models and larger-scale reinforcement learning.

4.2. Unsuccessful Attempts

In the early stages of developing DeepSeek-R1, we also encountered failures and setbacks along the way. We share our failure experiences here to provide insights, but this does not imply that these approaches are incapable of developing effective reasoning models.

Process Reward Model (PRM) PRM is a reasonable method to guide the model toward better approaches for solving reasoning tasks (Lightman et al., 2023; Uesato et al., 2022; Wang et al., 2023). However, in practice, PRM has three main limitations that may hinder its ultimate success. First, it is challenging to explicitly define a fine-grain step in general reasoning. Second, determining whether the current intermediate step is correct is a challenging task. Automated annotation using models may not yield satisfactory results, while manual annotation is not conducive to scaling up. Third, once a model-based PRM is introduced, it inevitably leads to reward hacking (Gao et al., 2022), and retraining the reward model needs additional training resources and it complicates the whole training pipeline. In conclusion, while PRM demonstrates a good ability to rerank the top-N responses generated by the model or assist in guided search (Snell et al., 2024), its advantages are limited compared to the additional computational overhead it introduces during the large-scale reinforcement learning process in our experiments.

Monte Carlo Tree Search (MCTS) Inspired by AlphaGo (Silver et al., 2017b) and AlphaZero (Silver et al., 2017a), we explored using Monte Carlo Tree Search (MCTS) to enhance test-time compute scalability. This approach involves breaking answers into smaller parts to allow the model to explore the solution space systematically. To facilitate this, we prompt the model to generate multiple tags that correspond to specific reasoning steps necessary for the search. For training, we first use collected prompts to find answers via MCTS guided by a pre-trained value model. Subsequently, we use the resulting question-answer pairs to train both the actor model and the value model, iteratively refining the process.

However, this approach encounters several challenges when scaling up the training. First, unlike chess, where the search space is relatively well-defined, token generation presents an

模型	AIME 2024		MATH-500	GPQA Diamond	LiveCodeBench
	pass@1	cons@64	pass@1	pass@1	pass@1
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6	55.0	40.2
DeepSeek-R1-Distill-Qwen-32B	72.6	83.3	94.3	62.1	57.2

Table 6 | 蒸馏和RL模型在推理相关基准测试上的比较。

RL训练, 达到了与QwQ-32B-Preview相当的性能。然而, 从DeepSeek-R1蒸馏而来的DeepSeek-R1-Distill-Qwen-32B, 在所有基准测试中都显著优于DeepSeek-R1-Zero-Qwen-32B。

因此, 我们可以得出两个结论: 第一, 将更强大的模型蒸馏到更小的模型中能取得优异的结果, 而依赖本文中提到的这种大规模RL的较小模型则需要巨大的计算能力, 甚至可能无法达到蒸馏的性能。第二, 虽然蒸馏策略既经济又有效, 但要超越智能的边界, 可能仍然需要更强大的基础模型和更大规模的强化学习。

4.2. 不成功的尝试

在开发DeepSeek-R1的早期阶段, 我们也遇到了过程中的失败和挫折。我们在此分享我们的失败经验以提供见解, 但这并不意味着这些方法无法开发出有效的推理模型。

过程奖励模型 (PRM) PRM 是一种合理的引导模型以更好地解决推理任务的方法 (Lightman 等人, 2023; Uesato 等人, 2022; Wang 等人, 2023)。然而, 在实践中, PRM 有三个主要限制可能会阻碍其最终的成功。首先, 在一般推理中明确定义一个细粒度步骤具有挑战性。其次, 确定当前中间步骤是否正确是一项具有挑战性的任务。使用模型进行自动标注可能无法获得令人满意的结果, 而手动标注不利于扩展。第三, 一旦引入基于模型的 PRM, 它不可避免地会导致奖励攻击 (Gao 等人, 2022), 并且重新训练奖励模型需要额外的训练资源, 并且会使整个训练流程复杂化。总之, 虽然 PRM 在重新排序模型生成的 Top-N 响应或协助引导搜索方面表现出良好的能力 (Snell 等人, 2024), 但与它在我们的实验中大规模强化学习过程中引入的额外计算开销相比, 其优势是有限的。

蒙特卡洛树搜索 (MCTS) 受 AlphaGo (Silver 等人, 2017b) 和 AlphaZero (Silver 等人, 2017a) 启发, 我们探索了使用蒙特卡洛树搜索 (MCTS) 来增强测试时的计算可扩展性。这种方法涉及将答案分解成更小的部分, 以允许模型系统地探索解决方案空间。为了促进这一点, 我们提示模型生成多个与特定推理步骤相对应的标签, 这些步骤对于搜索是必要的。对于训练, 我们首先使用收集到的提示通过 MCTS 找到答案, 并由预训练的值模型进行引导。随后, 我们使用生成的问答对来训练演员模型和值模型, 迭代地优化这个过程。

然而, 当扩展训练时, 这种方法会遇到一些挑战。首先, 与棋类游戏不同, 其中搜索空间相对明确, 而标记生成提出了一个

exponentially larger search space. To address this, we set a maximum extension limit for each node, but this can lead to the model getting stuck in local optima. Second, the value model directly influences the quality of generation since it guides each step of the search process. Training a fine-grained value model is inherently difficult, which makes it challenging for the model to iteratively improve. While AlphaGo's core success relied on training a value model to progressively enhance its performance, this principle proves difficult to replicate in our setup due to the complexities of token generation.

In conclusion, while MCTS can improve performance during inference when paired with a pre-trained value model, iteratively boosting model performance through self-search remains a significant challenge.

5. Conclusion, Limitations, and Future Work

In this work, we share our journey in enhancing model reasoning abilities through reinforcement learning. DeepSeek-R1-Zero represents a pure RL approach without relying on cold-start data, achieving strong performance across various tasks. DeepSeek-R1 is more powerful, leveraging cold-start data alongside iterative RL fine-tuning. Ultimately, DeepSeek-R1 achieves performance comparable to OpenAI-o1-1217 on a range of tasks.

We further explore distillation the reasoning capability to small dense models. We use DeepSeek-R1 as the teacher model to generate 800K training samples, and fine-tune several small dense models. The results are promising: DeepSeek-R1-Distill-Qwen-1.5B outperforms GPT-4o and Claude-3.5-Sonnet on math benchmarks with 28.9% on AIME and 83.9% on MATH. Other dense models also achieve impressive results, significantly outperforming other instruction-tuned models based on the same underlying checkpoints.

In the future, we plan to invest in research across the following directions for DeepSeek-R1.

- **General Capability:** Currently, the capabilities of DeepSeek-R1 fall short of DeepSeek-V3 in tasks such as function calling, multi-turn, complex role-playing, and JSON output. Moving forward, we plan to explore how long CoT can be leveraged to enhance tasks in these fields.
- **Language Mixing:** DeepSeek-R1 is currently optimized for Chinese and English, which may result in language mixing issues when handling queries in other languages. For instance, DeepSeek-R1 might use English for reasoning and responses, even if the query is in a language other than English or Chinese. We aim to address this limitation in future updates.
- **Prompting Engineering:** When evaluating DeepSeek-R1, we observe that it is sensitive to prompts. Few-shot prompting consistently degrades its performance. Therefore, we recommend users directly describe the problem and specify the output format using a zero-shot setting for optimal results.
- **Software Engineering Tasks:** Due to the long evaluation times, which impact the efficiency of the RL process, large-scale RL has not been applied extensively in software engineering tasks. As a result, DeepSeek-R1 has not demonstrated a huge improvement over DeepSeek-V3 on software engineering benchmarks. Future versions will address this by implementing rejection sampling on software engineering data or incorporating asynchronous evaluations during the RL process to improve efficiency.

指数级更大的搜索空间。为了解决这个问题，我们为每个节点设置了一个最大扩展限制，但这可能导致模型陷入局部最优。其次，值模型直接影响了生成质量，因为它指导了搜索过程的每一步。训练细粒度的值模型本质上很困难，这使得模型难以迭代地改进。虽然AlphaGo的核心成功依赖于训练值模型以逐步提升其性能，但由于标记生成的复杂性，这一原则在我们的设置中难以复制。

总之，虽然MCTS在与预训练的值模型配合时可以在推理过程中提高性能，但通过自我搜索迭代提升模型性能仍然是一个重大挑战。

5. 结论、局限性和未来工作

在这项工作中，我们分享了通过强化学习提升模型推理能力的历程。DeepSeek-R1-Zero代表了一种纯粹的RL方法，不依赖于冷启动数据，在各种任务中取得了优异的性能。

DeepSeek-R1更强大，利用冷启动数据并结合迭代RL微调。最终，DeepSeek-R1在一系列任务上的性能达到了与OpenAI-o1-1217相当的水平。

我们进一步探索了蒸馏推理能力到小密集模型。我们使用 DeepSeek-R1 作为教师模型来生成 800K 个训练样本，并微调几个小密集模型。结果令人鼓舞：

DeepSeek-R1-Distill-Qwen-1.5B 在数学基准测试中优于 GPT-4o 和 Claude-3.5-Sonnet，在 AIME 上达到 28.9%，在 MATH 上达到 83.9%。其他密集模型也取得了令人印象深刻的结果，显著优于基于相同底层检查点的其他指令微调模型。

未来，我们计划在 DeepSeek-R1 的以下方向进行投资研究。

- **通用能力：**目前，DeepSeek-R1 在函数调用、多轮、复杂角色扮演和 JSON 输出等任务中的能力不及 DeepSeek-V3。向前看，我们计划探索如何利用 CoT 来增强这些领域的任务。
- **语言混合：**DeepSeek-R1 目前针对中文和英文进行了优化，在处理其他语言查询时可能会导致语言混合问题。例如，即使查询不是英文或中文，DeepSeek-R1 也可能使用英文进行推理和响应。我们希望在未来的更新中解决这一限制。
- **提示工程：**在评估 DeepSeek-R1 时，我们观察到它对提示敏感。少样本提示会持续降低其性能。因此，我们建议用户直接描述问题，并在零样本设置中指定输出格式以获得最佳结果。
- **软件工程任务：**由于评估时间长，影响了 RL 进程的效率，大规模 RL 尚未在软件工程任务中得到广泛应用。因此，DeepSeek-R1 在软件工程基准测试上并未比 DeepSeek-V3 有巨大提升。未来版本将通过在软件工程数据上实施拒绝采样或在 RL 进程中结合异步评估来提高效率。

References

- AI@Meta. Llama 3.1 model card, 2024. URL https://github.com/meta-llama/llama-models/blob/main/models/llama3_1/MODEL_CARD.md.
- Anthropic. Claude 3.5 sonnet, 2024. URL <https://www.anthropic.com/news/claude-3-5-sonnet>.
- M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba. Evaluating large language models trained on code. *CoRR*, abs/2107.03374, 2021. URL <https://arxiv.org/abs/2107.03374>.
- A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Y. Dubois, B. Galambosi, P. Liang, and T. B. Hashimoto. Length-controlled alpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.
- X. Feng, Z. Wan, M. Wen, S. M. McAleer, Y. Wen, W. Zhang, and J. Wang. Alphazero-like tree-search can guide large language model decoding and training, 2024. URL <https://arxiv.org/abs/2309.17179>.
- L. Gao, J. Schulman, and J. Hilton. Scaling laws for reward model overoptimization, 2022. URL <https://arxiv.org/abs/2210.10760>.
- A. P. Gema, J. O. J. Leang, G. Hong, A. Devoto, A. C. M. Mancino, R. Saxena, X. He, Y. Zhao, X. Du, M. R. G. Madani, C. Barale, R. McHardy, J. Harris, J. Kaddour, E. van Krieken, and P. Minervini. Are we done with mmlu? *CoRR*, abs/2406.04127, 2024. URL <https://doi.org/10.48550/arXiv.2406.04127>.
- Google. Our next-generation model: Gemini 1.5, 2024. URL <https://blog.google/technology/ai/google-gemini-next-generation-model-february-2024>.
- Y. He, S. Li, J. Liu, Y. Tan, W. Wang, H. Huang, X. Bu, H. Guo, C. Hu, B. Zheng, et al. Chinese simpleqa: A chinese factuality evaluation for large language models. *arXiv preprint arXiv:2411.07140*, 2024.
- D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.
- Y. Huang, Y. Bai, Z. Zhu, J. Zhang, J. Zhang, T. Su, J. Liu, C. Lv, Y. Zhang, J. Lei, et al. C-Eval: A multi-level multi-discipline chinese evaluation suite for foundation models. *arXiv preprint arXiv:2305.08322*, 2023.
- N. Jain, K. Han, A. Gu, W. Li, F. Yan, T. Zhang, S. Wang, A. Solar-Lezama, K. Sen, and I. Stoica. Livecodebench: Holistic and contamination free evaluation of large language models for code. *CoRR*, abs/2403.07974, 2024. URL <https://doi.org/10.48550/arXiv.2403.07974>.

参考文献

- AI@Meta. Llama 3.1 model card, 2024. URL https://github.com/meta-llama/llama-models/blob/main/models/llama3_1/MODEL_CARD.md.
- Anthropic. Claude 3.5 sonnet, 2024. URL <https://www.anthropic.com/news/claude-3-5-sonnet>.
- M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, L. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba. Evaluating large language models trained on code. *CoRR*, abs/2107.03374, 2021. URL <https://arxiv.org/abs/2107.03374>.
- A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Y. Dubois, B. Galambosi, P. Liang, and T. B. Hashimoto. Length-controlled alpacaEval: A simple way to debias automatic evaluators. *arXiv preprint arXiv:2404.04475*, 2024.
- X. Feng, Z. Wan, M. Wen, S. M. McAleer, Y. Wen, W. Zhang, and J. Wang. Alphazero-like tree-search can guide large language model decoding and training, 2024. URL <https://arxiv.org/abs/2309.17179>.
- L. Gao, J. Schulman, and J. Hilton. Scaling laws for reward model overoptimization, 2022. URL <https://arxiv.org/abs/2210.10760>.
- A. P. Gema, J. O. J. Leang, G. Hong, A. Devoto, A. C. M. Mancino, R. Saxena, X. He, Y. Zhao, X. Du, M. R. G. Madani, C. Barale, R. McHardy, J. Harris, J. Kaddour, E. van Krieken, and P. Minervini. Are we done with mmlu? *CoRR*, abs/2406.04127, 2024. URL <https://doi.org/10.48550/arXiv.2406.04127>.
- Google. Our next-generation model: Gemini 1.5, 2024. URL <https://blog.google/technology/ai/google-gemini-next-generation-model-february-2024>.
- Y. He, S. Li, J. Liu, Y. Tan, W. Wang, H. Huang, X. Bu, H. Guo, C. Hu, B. Zheng, et al. Chinese simpleqa: A chinese factuality evaluation for large language models. *arXiv preprint arXiv:2411.07140*, 2024.
- D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.
- Y. Huang, Y. Bai, Z. Zhu, J. Zhang, J. Zhang, T. Su, J. Liu, C. Lv, Y. Zhang, J. Lei, et al. C-Eval: A multi-level multi-discipline chinese evaluation suite for foundation models. *arXiv preprint arXiv:2305.08322*, 2023.
- N. Jain, K. Han, A. Gu, W. Li, F. Yan, T. Zhang, S. Wang, A. Solar-Lezama, K. Sen, and I. Stoica. Livecodebench: Holistic and contamination free evaluation of large language models for code. *CoRR*, abs/2403.07974, 2024. URL <https://doi.org/10.48550/arXiv.2403.07974>.

S. Krishna, K. Krishna, A. Mohananey, S. Schwarcz, A. Stambler, S. Upadhyay, and M. Faruqui. Fact, fetch, and reason: A unified evaluation of retrieval-augmented generation. *CoRR*, abs/2409.12941, 2024. doi: 10.48550/ARXIV.2409.12941. URL <https://doi.org/10.48550/arXiv.2409.12941>.

A. Kumar, V. Zhuang, R. Agarwal, Y. Su, J. D. Co-Reyes, A. Singh, K. Baumli, S. Iqbal, C. Bishop, R. Roelofs, et al. Training language models to self-correct via reinforcement learning. *arXiv preprint arXiv:2409.12917*, 2024.

H. Li, Y. Zhang, F. Koto, Y. Yang, H. Zhao, Y. Gong, N. Duan, and T. Baldwin. CMMLU: Measuring massive multitask language understanding in Chinese. *arXiv preprint arXiv:2306.09212*, 2023.

T. Li, W.-L. Chiang, E. Frick, L. Dunlap, T. Wu, B. Zhu, J. E. Gonzalez, and I. Stoica. From crowdsourced data to high-quality benchmarks: Arena-hard and benchbuilder pipeline. *arXiv preprint arXiv:2406.11939*, 2024.

H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

B. Y. Lin. ZeroEval: A Unified Framework for Evaluating Language Models, July 2024. URL <https://github.com/WildEval/ZeroEval>.

MAA. American invitational mathematics examination - aime. In *American Invitational Mathematics Examination - AIME 2024*, February 2024. URL <https://maa.org/math-competitions/american-invitational-mathematics-examination-aime>.

OpenAI. Hello GPT-4o, 2024a. URL <https://openai.com/index/hello-gpt-4o/>.

OpenAI. Learning to reason with llms, 2024b. URL <https://openai.com/index/learning-to-reason-with-llms/>.

OpenAI. Introducing SimpleQA, 2024c. URL <https://openai.com/index/introducing-simpleqa/>.

OpenAI. Introducing SWE-bench verified we’re releasing a human-validated subset of swe-bench that more, 2024d. URL <https://openai.com/index/introducing-swe-bench-verified/>.

Qwen. Qwq: Reflect deeply on the boundaries of the unknown, 2024a. URL <https://qwenlm.github.io/blog/qwq-32b-preview/>.

Qwen. Qwen2.5: A party of foundation models, 2024b. URL <https://qwenlm.github.io/blog/qwen2.5>.

D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman. GPQA: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*, 2023.

Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, M. Zhang, Y. Li, Y. Wu, and D. Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. P. Lillicrap, K. Simonyan, and D. Hassabis. Mastering chess and shogi by self-play with a general reinforcement learning algorithm. *CoRR*, abs/1712.01815, 2017a. URL <http://arxiv.org/abs/1712.01815>.

S. Krishna, K. Krishna, A. Mohananey, S. Schwarcz, A. Stambler, S. Upadhyay, and M. Faruqui. Fact, fetch, and reason: A unified evaluation of retrieval-augmented generation. *CoRR*, abs/2409.12941, 2024. doi: 10.48550/ARXIV.2409.12941. URL <https://doi.org/10.48550/arXiv.2409.12941>.

A. Kumar, V. Zhuang, R. Agarwal, Y. Su, J. D. Co-Reyes, A. Singh, K. Baumli, S. Iqbal, C. Bishop, R. Roelofs, et al. Training language models to self-correct via reinforcement learning. *arXiv preprint arXiv:2409.12917*, 2024.

H. Li, Y. Zhang, F. Koto, Y. Yang, H. Zhao, Y. Gong, N. Duan, and T. Baldwin. CMMLU: Measuring massive multitask language understanding in Chinese. *arXiv preprint arXiv:2306.09212*, 2023.

T. Li, W.-L. Chiang, E. Frick, L. Dunlap, T. Wu, B. Zhu, J. E. Gonzalez, and I. Stoica. From crowdsourced data to high-quality benchmarks: Arena-hard and benchbuilder pipeline. *arXiv preprint arXiv:2406.11939*, 2024.

H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

B. Y. Lin. ZeroEval: A Unified Framework for Evaluating Language Models, July 2024. URL <https://github.com/WildEval/ZeroEval>.

MAA. American invitational mathematics examination - aime. In *American Invitational Mathematics Examination - AIME 2024*, February 2024. URL <https://maa.org/math-competitions/american-invitational-mathematics-examination-aime>.

OpenAI. Hello GPT-4o, 2024a. URL <https://openai.com/index/hello-gpt-4o/>.

OpenAI. Learning to reason with llms, 2024b. URL <https://openai.com/index/learning-to-reason-with-llms/>.

OpenAI. Introducing SimpleQA, 2024c. URL <https://openai.com/index/introducing-simpleqa/>.

OpenAI. Introducing SWE-bench verified we’re releasing a human-validated subset of swe-bench that more, 2024d. URL <https://openai.com/index/introducing-swe-bench-verified/>.

Qwen. Qwq: Reflect deeply on the boundaries of the unknown, 2024a. URL <https://qwenlm.github.io/blog/qwq-32b-preview/>.

Qwen. Qwen2.5: A party of foundation models, 2024b. URL <https://qwenlm.github.io/blog/qwen2.5>.

D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman. GPQA: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*, 2023.

Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, M. Zhang, Y. Li, Y. Wu, and D. Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. P. Lillicrap, K. Simonyan, and D. Hassabis. Mastering chess and shogi by self-play with a general reinforcement learning algorithm. *CoRR*, abs/1712.01815, 2017a. URL <http://arxiv.org/abs/1712.01815>.

- D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, Y. Chen, T. P. Lillicrap, F. Hui, L. Sifre, G. van den Driessche, T. Graepel, and D. Hassabis. Mastering the game of go without human knowledge. *Nat.*, 550(7676):354–359, 2017b. doi: 10.1038/NATURE24270. URL <https://doi.org/10.1038/nature24270>.
- C. Snell, J. Lee, K. Xu, and A. Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters, 2024. URL <https://arxiv.org/abs/2408.03314>.
- T. Trinh, Y. Wu, Q. Le, H. He, and T. Luong. Solving olympiad geometry without human demonstrations. *Nature*, 2024. doi: 10.1038/s41586-023-06747-5.
- J. Uesato, N. Kushman, R. Kumar, F. Song, N. Siegel, L. Wang, A. Creswell, G. Irving, and I. Higgins. Solving math word problems with process-and outcome-based feedback. *arXiv preprint arXiv:2211.14275*, 2022.
- P. Wang, L. Li, Z. Shao, R. Xu, D. Dai, Y. Li, D. Chen, Y. Wu, and Z. Sui. Math-shepherd: A label-free step-by-step verifier for llms in mathematical reasoning. *arXiv preprint arXiv:2312.08935*, 2023.
- X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arulraj, X. He, Z. Jiang, T. Li, M. Ku, K. Wang, A. Zhuang, R. Fan, X. Yue, and W. Chen. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *CoRR*, abs/2406.01574, 2024. URL <https://doi.org/10.48550/arXiv.2406.01574>.
- C. S. Xia, Y. Deng, S. Dunn, and L. Zhang. Agentless: Demystifying llm-based software engineering agents. *arXiv preprint*, 2024.
- H. Xin, Z. Z. Ren, J. Song, Z. Shao, W. Zhao, H. Wang, B. Liu, L. Zhang, X. Lu, Q. Du, W. Gao, Q. Zhu, D. Yang, Z. Gou, Z. F. Wu, F. Luo, and C. Ruan. Deepseek-prover-v1.5: Harnessing proof assistant feedback for reinforcement learning and monte-carlo tree search, 2024. URL <https://arxiv.org/abs/2408.08152>.
- J. Zhou, T. Lu, S. Mishra, S. Brahma, S. Basu, Y. Luan, D. Zhou, and L. Hou. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

- D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, Y. Chen, T. P. Lillicrap, F. Hui, L. Sifre, G. van den Driessche, T. Graepel, and D. Hassabis. Mastering the game of go without human knowledge. *Nat.*, 550(7676):354–359, 2017b. doi: 10.1038/NATURE24270. URL <https://doi.org/10.1038/nature24270>.
- C. Snell, J. Lee, K. Xu, and A. Kumar. Scaling llm test-time compute optimally can be more effective than scaling model parameters, 2024. URL <https://arxiv.org/abs/2408.03314>.
- T. Trinh, Y. Wu, Q. Le, H. He, and T. Luong. Solving olympiad geometry without human demonstrations. *Nature*, 2024. doi: 10.1038/s41586-023-06747-5.
- J. Uesato, N. Kushman, R. Kumar, F. Song, N. Siegel, L. Wang, A. Creswell, G. Irving, and I. Higgins. Solving math word problems with process-and outcome-based feedback. *arXiv preprint arXiv:2211.14275*, 2022.
- P. Wang, L. Li, Z. Shao, R. Xu, D. Dai, Y. Li, D. Chen, Y. Wu, and Z. Sui. Math-shepherd: A label-free step-by-step verifier for llms in mathematical reasoning. *arXiv preprint arXiv:2312.08935*, 2023.
- X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*, 2022.
- Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arulraj, X. He, Z. Jiang, T. Li, M. Ku, K. Wang, A. Zhuang, R. Fan, X. Yue, and W. Chen. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *CoRR*, abs/2406.01574, 2024. URL <https://doi.org/10.48550/arXiv.2406.01574>.
- C. S. Xia, Y. Deng, S. Dunn, and L. Zhang. Agentless: Demystifying llm-based software engineering agents. *arXiv preprint*, 2024.
- H. Xin, Z. Z. Ren, J. Song, Z. Shao, W. Zhao, H. Wang, B. Liu, L. Zhang, X. Lu, Q. Du, W. Gao, Q. Zhu, D. Yang, Z. Gou, Z. F. Wu, F. Luo, and C. Ruan. Deepseek-prover-v1.5: Harnessing proof assistant feedback for reinforcement learning and monte-carlo tree search, 2024. URL <https://arxiv.org/abs/2408.08152>.
- J. Zhou, T. Lu, S. Mishra, S. Brahma, S. Basu, Y. Luan, D. Zhou, and L. Hou. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

Appendix

A. Contributions and Acknowledgments

Core Contributors

Daya Guo
Dejian Yang
Haowei Zhang
Junxiao Song
Ruoyu Zhang
Runxin Xu
Qihao Zhu
Shirong Ma
Peiyi Wang
Xiao Bi
Xiaokang Zhang
Xingkai Yu
Yu Wu
Z.F. Wu
Zhibin Gou
Zhihong Shao
Zhuoshu Li
Ziyi Gao

Contributors

Aixin Liu
Bing Xue
Bingxuan Wang
Bochao Wu
Bei Feng
Chengda Lu
Chenggang Zhao
Chengqi Deng
Chong Ruan
Damai Dai
Deli Chen
Dongjie Ji
Erhang Li
Fangyun Lin
Fucong Dai
Fuli Luo*
Guangbo Hao
Guanting Chen
Guowei Li
H. Zhang
Hanwei Xu
Honghui Ding
Huazuo Gao
Hui Qu

Hui Li
Jianzhong Guo
Jiashi Li
Jingchang Chen
Jingyang Yuan
Jinhao Tu
Junjie Qiu
Junlong Li
J.L. Cai
Jiaqi Ni
Jian Liang
Jin Chen
Kai Dong
Kai Hu*
Kaichao You
Kaige Gao
Kang Guan
Kexin Huang
Kuai Yu
Lean Wang
Lecong Zhang
Liang Zhao
Litong Wang
Liyue Zhang
Lei Xu
Leyi Xia
Mingchuan Zhang
Minghua Zhang
Minghui Tang
Mingxu Zhou
Meng Li
Miaojun Wang
Mingming Li
Ning Tian
Panpan Huang
Peng Zhang
Qiancheng Wang
Qinyu Chen
Qiushi Du
Ruiqi Ge*
Ruisong Zhang
Ruizhe Pan
Runji Wang
R.J. Chen
R.L. Jin

Appendix

A. Contributions and Acknowledgments

核心贡献者

Daya Guo
Dejian Yang
Haowei Zhang
Junxiao Song
Ruoyu Zhang
Runxin Xu
Qihao Zhu
Shirong Ma
Peiyi Wang
Xiao Bi
Xiaokang Zhang
Xingkai Yu
Yu Wu
Z.F. Wu
Zhibin Gou
Zhihong Shao
Zhuoshu Li
Ziyi Gao

贡献者

Aixin Liu
Bing Xue
Bingxuan Wang
Bochao Wu
Bei Feng
Chengda Lu
Chenggang Zhao
Chengqi Deng
Chong Ruan
Damai Dai
Deli Chen
Dongjie Ji
Erhang Li
Fangyun Lin
Fucong Dai
Fuli Luo*
Guangbo Hao
Guanting Chen
Guowei Li
H. Zhang
Hanwei Xu
Honghui Ding
Huazuo Gao
Hui Qu

惠立建
忠国级史景
长陈景阳
元金浩涂
俊杰邱俊龙
李J.
L.蔡嘉琪
倪健梁金
陈凯东
凯胡*凯超
尤凯高康
关克欣黄
快余Lean
王乐丛
张亮赵立通
王立月
张雷徐乐怡
夏明川
张明华
张明辉唐明
旭周梦李妙军
王明明
李宁天盼盼
黄鹏
张千成王庆余
陈秋石杜瑞琪
郭*瑞松
张瑞哲潘润基
王R.J.陈R.L.金

Ruyi Chen
Shanghao Lu
Shangyan Zhou
Shanhuang Chen
Shengfeng Ye
Shiyu Wang
Shuiping Yu
Shunfeng Zhou
Shuting Pan
S.S. Li
Shuang Zhou
Shaoqing Wu
Shengfeng Ye
Tao Yun
Tian Pei
Tianyu Sun
T. Wang
Wangding Zeng
Wen Liu
Wenfeng Liang
Wenjun Gao
Wenqin Yu*
Wentao Zhang
W.L. Xiao
Wei An
Xiaodong Liu
Xiaohan Wang
Xiaokang Chen
Xiaotao Nie
Xin Cheng
Xin Liu
Xin Xie
Xingchao Liu
Xinyu Yang
Xinyuan Li
Xuecheng Su
Xuheng Lin
X.Q. Li
Xiangyue Jin
Xiaojin Shen
Xiaosha Chen
Xiaowen Sun
Xiaoxiang Wang
Xinnan Song
Xinyi Zhou
Xianzu Wang
Xinxia Shan
Y.K. Li
Y.Q. Wang

Y.X. Wei
Yang Zhang
Yanhong Xu
Yao Li
Yao Zhao
Yaofeng Sun
Yaohui Wang
Yi Yu
Yichao Zhang
Yifan Shi
Yiliang Xiong
Ying He
Yishi Piao
Yisong Wang
Yixuan Tan
Yiyang Ma*
Yiyuan Liu
Yongqiang Guo
Yuan Ou
Yuduan Wang
Yue Gong
Yuheng Zou
Yujia He
Yunfan Xiong
Yuxiang Luo
Yuxiang You
Yuxuan Liu
Yuyang Zhou
Y.X. Zhu
Yanping Huang
Yaohui Li
Yi Zheng
Yuchen Zhu
Yunxian Ma
Ying Tang
Yukun Zha
Yuting Yan
Z.Z. Ren
Zehui Ren
Zhangli Sha
Zhe Fu
Zhean Xu
Zhenda Xie
Zhengyan Zhang
Zhewen Hao
Zhicheng Ma
Zhigang Yan
Zhiyu Wu
Zihui Gu

陈瑞怡 上海路尚
岩 周山皇 陈生风
叶时雨 王淑萍 余
顺峰 周淑婷 潘
S.S. 李双 周少庆
吴生风 叶涛 云天
鲍天宇 孙 T. 王
王定 曾文 刘文锋
梁文俊 高文钦 余
* 文涛 张W.L. 肖
伟 安晓东 刘晓寒
王晓康 陈晓涛 聂
欣 成欣 刘欣 谢
兴超 刘新宇 杨新
元 李学成 苏旭恒
林 X.Q. 李向月
金晓金 沈晓莎 陈
晓文 孙晓祥 王新
南 宋新怡 周先祖
王新霞 山 Y.K.
李Y.Q. 王

魏Y.X. 魏杨 张艳
红 徐瑶 李瑶 赵
瑶峰 孙瑶辉 王一
余一超 张一帆 石
一良 雄英 何一石
骑一松 王一轩 谭
一阳 马* 一元 刘
永强 郭园 欧宇段
王越 龚雨恒 邹宇
嘉 何云帆 雄宇翔
罗宇翔 尤宇轩 刘
宇阳 周宇翔 朱
Y.X. 朱艳平 黄瑶
辉 李一 郑宇晨
朱云仙 马英 唐雨
坤 赵宇婷 严
Z.Z. 任泽辉 任张
丽 沙哲 付哲 安
徐振达 谢正岩 张
正文 郝志成 马志
刚 岳志宇 吴志辉

Zijia Zhu
Zijun Liu*
Zilin Li
Ziwei Xie
Ziyang Song
Zizheng Pan

Zhen Huang
Zhipeng Xu
Zhongyu Zhang
Zhen Zhang

Within each role, authors are listed alphabetically by the first name. Names marked with * denote individuals who have departed from our team.

Zijia Zhu
Zijun Liu*
Zilin Li
Ziwei Xie
Ziyang Song
Zizheng Pan

Zhen Huang
Zhipeng Xu
Zhongyu Zhang
Zhen Zhang

在每个角色中，作者按名字首字母排序列出。带*标记的名字表示已离开我们团队的成员。